

Join or Wait? Seller Capability Development and Dynamic Platform Pricing*

Kevin Mi Danxia Xie Buyuan Yang Hanzhe Zhang[†]

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Abstract

Platforms use fee concessions to recruit sellers and costly support to build their capability. We develop a two-period model of a monopoly platform in which heterogeneous sellers choose when to enter, only early entrants acquire capability, and the platform cannot commit to future fees. Future entrant fees determine the value of waiting and hence the concession required for early entry. As the return to early capability rises, the unique limited-commitment equilibrium passes through four regimes: positive late entry, a zero-late-entry plateau, renewed early recruitment, and seller saturation. The plateau arises because recruiting an additional early seller lowers the waiting concession paid to the entire early cohort, a saving that disappears when late entry closes. At that boundary, the marginal early seller would earn no rent by waiting, yet the platform still subsidizes early entry to expand buyer-side revenue. With active late entry, greater capability front-loads seller entry yet expands period-2 participation on both sides. When the platform and sellers discount the future at the same rate, full commitment eliminates late entry but can reduce early recruitment. Whenever allocations differ across commitment regimes, full commitment raises platform profit while reducing period-2 participation on both sides and lowering social welfare, even though support intensity is unchanged. Thus, resolving the platform's time-inconsistency problem can benefit the platform at the expense of market development.

Keywords: two-sided markets; seller capability; limited commitment; dynamic pricing.

JEL Classification: D42, D62, L14, L86.

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[†]Mi: Department of Economics, Michigan State University (E-mail: kwmi@msu.edu); Xie: Institute of Economics, School of Social Sciences, Tsinghua University (E-mail: xiedanxia@tsinghua.edu.cn); Yang: School of International Trade and Economics, Central University of Finance and Economics (E-mail: yangbuyuan@cufe.edu.cn); Zhang: Department of Economics, Michigan State University (E-mail: hanzhe@msu.edu).

1 Introduction

Digital marketplaces build their supply sides through both acquisition pricing and real investment. Platforms can use fee concessions or direct transfers to recruit third-party providers and can invest in integration tools that reduce providers’ development costs (Tan et al., 2020). Ancillary services such as logistics and order fulfillment can likewise lower operating costs or relax seller capacity constraints (Li et al., 2023). These instruments are economically distinct: acquisition transfers redistribute surplus, whereas capability investment uses real resources and raises the future value generated by sellers who enter early.

Sellers choose whether to enter early, contribute to current buyer demand, and acquire capability, or instead wait until the market develops. Once early entry is realized, a platform that lacks commitment reoptimizes the late-entrant fee without internalizing how those future terms affect the ex ante value of waiting and hence the cost of early recruitment. We ask how the platform should combine acquisition pricing and capability investment when limited commitment over future fees shapes sellers’ incentive to wait. Does full commitment expand market development, or can it instead raise platform profit while reducing future participation and social welfare?

China’s rural e-commerce expansion illustrates these platform choices. Alibaba’s Rural Taobao program combined a county-and-village service network with logistics assistance, e-commerce training, and rural financial services. By the end of 2018, the program covered 1,174 counties and included more than 30,000 village-level service stations (World Bank and Alibaba Group, 2019, pp. 55–58). Figure 1 documents the broader expansion of rural online retailing and Taobao Villages, both nationally and in counties designated as poor during the sample period. Taobao Villages are rural communities that meet minimum thresholds for active online shops and annual online sales (World Bank and Alibaba Group, 2019, pp. 102–103). These institutional features motivate the platform’s three strategic choices in our model: early seller recruitment, capability investment, and continuation pricing.

We develop a two-period model of a monopoly platform with heterogeneous, long-lived sellers and a new cohort of heterogeneous, short-lived buyers in each period. Cross-side benefits link participation on the two sides. In period 1, the platform sets access fees for buyers and sellers and commits to costly support for early sellers, but it cannot commit to period-2 fees. Sellers

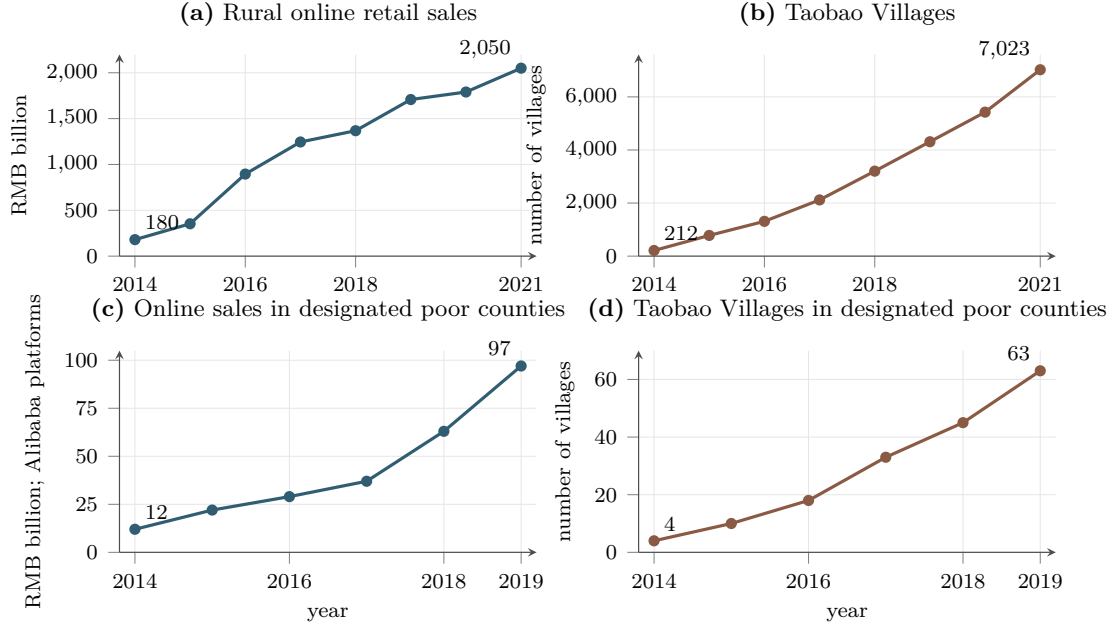


Figure 1: China’s Rural E-Commerce and Taobao Village Expansion, 2014–2021

Notes. Panels (a) and (b) report the published national rural-sales series and the number of Taobao Villages. Panels (c) and (d) report Alibaba-platform seller sales and Taobao Villages for the 832 counties designated under China’s national poverty-alleviation program. Sales are nominal RMB billions. Panel (a) is not a constant-definition series: the 2014–2015 observations are rural online-shopping or transaction estimates, whereas the 2016–2021 observations are Ministry of Commerce seller-location rural online retail sales. In panel (c), the source disclosures for 2014, 2017, and 2018 are lower bounds (“more than” RMB 12, 37, and 63 billion); the figure places those observations at the disclosed thresholds. The 2015 and 2019 source values, RMB 21.556 and 97.4 billion, are displayed as 22 and 97. The county sales series covers Alibaba platforms and is not comparable in level with the national series. The 2020 village count in panel (b) applies AliResearch’s revised rule that counts agricultural-product sales twice toward the sales threshold. The county panels end in 2019 because all 832 counties exited the designation in 2020. Sources include China’s Ministry of Commerce ([Ministry of Commerce of the People’s Republic of China, 2017, 2020, 2022b,a](#)), AliResearch ([AliResearch, 2015](#)), [World Bank and Alibaba Group \(2019\)](#), and [Huang \(2020\)](#). Online Appendix 1.1 gives the value-by-value source audit and live links.

differ in a recurring operating friction and choose whether to enter early or wait. Only early sellers acquire capability through learning by doing and platform support, which raises their gross participation value in period 2. In period 2, the platform sets the buyer fee and separate seller fees for incumbents and late entrants. Sellers therefore compare the lifetime payoff from entering early with the payoff from waiting. Continuation pricing thus feeds back into the current cost of early recruitment, while acquired capability creates a cohort-specific return that the platform can monetize ex post. To isolate this commitment problem, we compare the limited-commitment game with a full-commitment benchmark in which the platform commits, before period-1 entry, to a complete history-contingent schedule of continuation fees.

Our analysis yields four main findings. The first three characterize limited commitment, and the fourth compares limited and full commitment. First, the limited-commitment game has a unique on-path allocation. Every period-1 seller remains active in period 2, and the platform provides positive support because acquired capability can be monetized through the incumbent-seller fee. As the net return to early capability rises, the equilibrium moves from positive late entry to a zero-late-entry plateau, then to renewed early recruitment without late entry, and finally to seller saturation. The plateau is generated by a kink in early-acquisition incentives. Before late entry closes, an additional early seller reduces both subsequent entry and the waiting concession embedded in the common fee paid to the entire early cohort. At closure, this cohort-wide marginal saving disappears, locally pinning early recruitment.

Second, early-seller subsidies are not solely compensation for forgoing the option to wait. The platform can charge a positive early-seller fee only while late entry remains active and the return to early capability is sufficiently low. As capability becomes more valuable, the platform recruits higher-friction sellers earlier, pushing the fee downward. The accompanying decline in late entry reduces the waiting concession, but this offset is smaller than the increase in the marginal seller's operating friction. The fee therefore becomes negative before late entry disappears. At the entry-closure boundary, the waiting concession is already zero, yet the platform continues to subsidize early sellers because they generate seller-to-buyer revenue.

Third, capability and seller patience affect market development through distinct intertemporal margins. With positive late entry, greater capability expands early recruitment by more than it displaces late entry, so the period-2 seller base grows. Buyer participation rises in both periods, more strongly in period 1. Seller patience raises the value of waiting and hence the direct cost of early recruitment, but it also increases the cohort-wide saving from reducing the waiting concession. The first force dominates when the early cohort is small; the second dominates once the cohort exceeds half the seller mass required to eliminate late entry. Seller patience can therefore either delay or accelerate market development.

Finally, full commitment eliminates late entry by allowing the platform to choose future entrant terms before sellers decide when to enter. Relative to limited commitment, it induces more early sellers when capability returns are low, fewer at intermediate levels, and the same number once capability returns are sufficiently high. Whenever the regimes induce different allocations, full

commitment raises platform profit but reduces period-2 participation on both sides and lowers social welfare. The profit gain comes from internalizing how future entrant terms affect the current value of waiting. Because the two regimes choose the same support intensity, the welfare ranking is instead driven by participation: the larger period-2 market under limited commitment generates cross-side benefits that exceed the operating frictions of the additional sellers. Full commitment thereby resolves the platform’s time-inconsistency problem at the expense of market development and social welfare.

In the remainder of this section, we place this work within the related literature. Section 2 presents the model, and Section 3 characterizes the limited-commitment equilibrium. Section 4 compares limited and full commitment regarding market development, platform profit, and social welfare. Section 5 discusses empirical and managerial implications, and Section 6 concludes. The Online Appendix contains all proofs and supplementary analyses.

Related Literature. Our paper contributes to four strands of literature. First, our study enriches the literature on dynamic platform development and price commitment. Canonical models of two-sided markets show how cross-side effects shape participation and the structure of platform prices (Rochet and Tirole, 2003; Parker and Van Alstyne, 2005; Armstrong, 2006; Rochet and Tirole, 2006; Weyl, 2010), while the literature on platform launch examines adoption coordination (Caillaud and Jullien, 2003). Hagiu (2006) provides a particularly close commitment benchmark: sellers arrive before buyers, and the platform chooses whether to commit to the buyer price when announcing the seller price. Related studies examine intertemporal platform membership (Cabral, 2019), the joint determination of platform adoption and affiliated suppliers’ entry and pricing (Zhou, 2017), and launch strategies based on piggybacking on external networks or token issuance (Dou and Wu, 2021; Bakos and Halaburda, 2022). In contrast, we endogenize the timing of seller entry when early participation creates cohort-specific capability and continuation pricing determines the ex ante value of waiting—and hence the concession required to induce early entry.

Second, our work contributes to the literature on platform investment. Existing studies analyze investment in core platform performance under network externalities (Anderson et al., 2014), the joint determination of two-sided prices and costly integration investment (Tan et al., 2020), and pricing and engineering investment over the platform life cycle (Anderson et al., 2023). Related work considers ancillary services that relax seller-side capacity constraints but may crowd out sellers’ own

investment (Li et al., 2023). Classic models of learning by doing show how production experience can increase future productivity or reduce unit costs (Arrow, 1962; Spence, 1981). In contrast, we connect these ideas by distinguishing real support expenditure from acquisition transfers and by allowing only early sellers to acquire capability. The platform’s return to support consequently depends on both when sellers enter and how it subsequently prices incumbent sellers.

Third, our paper contributes to the literature on history-based pricing. Villas-Boas (2004) studies a monopolist that charges different prices to previous and new customers when consumers are forward-looking. Acquisti and Varian (2005) examine pricing contingent on purchase history, including settings with enhanced services and limited commitment. The literature on two-sided markets also studies targeted pricing, versioning, and screening across observable groups or privately informed types (Liu and Serfes, 2013; Lin, 2020; Jeon et al., 2022; de Cornière et al., 2025). In contrast, participation history in our model reflects not only past behavior but also a productive difference: the capability acquired through early entry. History-contingent pricing therefore enables the platform to monetize capability formation, rather than serving solely as an instrument for behavior-based price discrimination or screening.

Finally, our motivating application connects the paper to the empirical literature on Chinese e-commerce, particularly in rural and geographically remote markets. Existing work quantifies the spatial welfare and distributional effects of e-commerce-induced reductions in trade costs (Fan et al., 2018) and evaluates a rural e-commerce expansion whose gains arose primarily from overcoming logistical barriers (Couture et al., 2021). Related studies examine selection into rural e-commerce and associated income differences (Luo and Niu, 2019), the formation of Taobao Villages (Qi et al., 2019), and conditions relating to human capital, infrastructure, logistics, and the business environment (World Bank and Alibaba Group, 2019). Evidence from Taobao’s broader marketplace also documents large and asymmetric buyer–seller network effects (Chu and Manchanda, 2016). This empirical literature motivates our focus on market access, complementary inputs, and cross-side market development. Our theory complements this work by characterizing how capability support and cohort pricing jointly shape the timing and scale of seller participation.

2 The Model

2.1 Economic Environment

A monopoly platform operates a marketplace over two periods, $t \in \{1, 2\}$. The platform intermediates between a unit mass of sellers that may participate in either or both periods and a new unit mass of short-lived buyers in each period. Each seller draws a platform-operating friction $\theta_i \sim \text{Unif}[0, 1]$ once before the period-1 participation decision, observes it privately, and retains the same type in period 2. Each short-lived buyer independently draws a stand-alone participation value $v_{jt} \sim \text{Unif}[0, 1]$ upon arrival and observes it privately. All draws are independent across agents and, for buyers, across periods. An active seller incurs θ_i in each period of participation. Thus, θ_i captures a recurring friction associated with using digital tools, complying with platform rules, and managing online operations rather than a one-time adoption cost.

Let n_t^b and n_t^s denote the masses of participating buyers and sellers, respectively, in period t . In period 2, let n_{2I}^s denote the mass of period-1 sellers that remain active, and let n_{2E}^s denote the mass of sellers that wait in period 1 and first participate in period 2. Thus, we have

$$0 \leq n_{2I}^s \leq n_1^s, \quad 0 \leq n_{2E}^s \leq 1 - n_1^s, \quad \text{and} \quad n_2^s = n_{2I}^s + n_{2E}^s.$$

We refer to period-1 sellers who remain active in period 2 as incumbents and to sellers who first become active in period 2 as late entrants. An active seller receives a cross-side benefit αn_t^b from buyer participation, while an active buyer receives a cross-side benefit βn_t^s from seller participation, where $\alpha, \beta > 0$. Participation is a binary choice in each period and denotes active platform membership rather than account registration. Utilities are quasi-linear, and nonparticipation yields zero. The platform charges a buyer fee p_t^b in period t , a period-1 seller fee p_1^s , and period-2 seller fees p_{2I}^s and p_{2E}^s . The latter two fees apply to incumbents and late entrants, respectively. Fees are publicly posted for the relevant participation-history group and may be negative. The platform cannot reject a seller who accepts the applicable posted fee. Accordingly, $n_{2E}^s = 0$ means that no positive mass of waiting sellers enters under the equilibrium terms; it does not mean that the platform closes registration or otherwise denies access.

A seller who participates in period 1 becomes more capable in period 2 through two channels: an exogenous learning-by-doing increment $c \geq 0$ and platform support $x \geq 0$. The platform chooses and funds support in period 1. Support is delivered after period-1 participation and before period-2 pricing, so its cost is sunk when continuation fees are chosen. After this one-period lag, the combined capability increment $c + x$ raises the period-2 payoff of every period-1 seller. Providing support costs $(\kappa x^2/2)n_1^s$, where $\kappa > 0$ governs cost curvature and κx is the marginal resource cost of support per early seller. The exogenous component c entails no platform expenditure. Apart from support, the platform has zero operating costs. Support is a real resource cost, whereas access fees, including negative fees, are transfers.

Let u_{i1}^s denote seller i 's period-1 utility. In period 2, let $u_{i2,I}^s$ denote its utility as an incumbent and $u_{i2,E}^s$ its utility as a late entrant. Let u_{jt}^b denote buyer j 's utility in period t . Conditional on active participation, the seller and buyer flow utilities are

$$u_{i1}^s = \alpha n_1^b - \theta_i - p_1^s, \quad (1)$$

$$u_{i2,I}^s = \alpha n_2^b - \theta_i + c + x - p_{2I}^s, \quad (2)$$

$$u_{i2,E}^s = \alpha n_2^b - \theta_i - p_{2E}^s, \quad (3)$$

$$u_{jt}^b = v_{jt} + \beta n_t^s - p_t^b, \quad t = 1, 2. \quad (4)$$

Sellers discount period-2 utility by $\delta_s \in (0, 1)$. A seller's intertemporal utility is the discounted sum of its realized flow utilities. The platform discounts period-2 profit by $\delta_p \in (0, 1]$. Sellers face no liquidity constraints.

2.2 Timing and Limited Commitment

In period 1, the platform posts (p_1^b, p_1^s) and commits to provide support x to every seller who participates in period 1. Crucially, the platform cannot commit to the period-2 fees $(p_2^b, p_{2I}^s, p_{2E}^s)$, which are chosen only after period-1 participation histories become observable. Figure 2 summarizes the timing of the game.

Within each period, buyers and eligible sellers make participation decisions simultaneously after observing the applicable fees and their own types. Period-1 sellers anticipate the continuation equilibrium when deciding whether to join. At the end of period 1, the aggregate early-seller mass

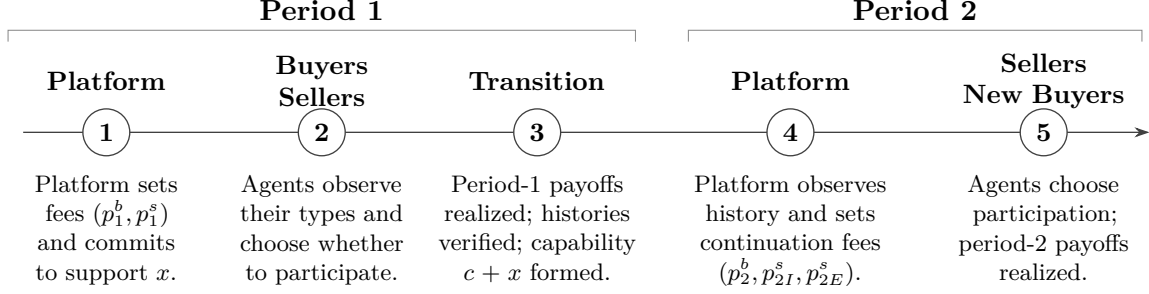


Figure 2: Timing of the Game

and individual seller participation histories become publicly observable and verifiable. The platform can then distinguish incumbents from sellers that waited, and all period-2 agents condition their decisions on the same inherited state.

Given the inherited seller mass n_1^s and support level x , the platform's period-2 continuation value is

$$\Pi_2^*(n_1^s, x) = \max_{p_2^b, p_{2I}^s, p_{2E}^s} \left\{ p_2^b n_2^b + p_{2I}^s n_{2I}^s + p_{2E}^s n_{2E}^s \right\}.$$

Anticipating this sequentially optimal continuation policy, the platform solves

$$\Pi^* = \max_{p_1^b, p_1^s, x \geq 0} \left\{ p_1^b n_1^b + p_1^s n_1^s - \frac{\kappa x^2}{2} n_1^s + \delta_p \Pi_2^*(n_1^s, x) \right\}.$$

In both problems, participation masses are those induced by the applicable fees and agents' participation decisions. We next characterize these decisions and define the equilibrium.

2.3 Participation Strategies and Equilibrium

Fix a period-1 participation history and the continuation equilibrium that follows it. The continuation values available to a seller after entering early and after waiting are, respectively,

$$\begin{aligned} V_{2I}^s(\theta) &= \max \left\{ 0, \alpha n_2^b - \theta + c + x - p_{2I}^s \right\}, \\ V_{2E}^s(\theta) &= \max \left\{ 0, \alpha n_2^b - \theta - p_{2E}^s \right\}. \end{aligned}$$

The additional utility that seller θ gains from participating in period 1 rather than waiting is

$$\Delta U_1^s(\theta) = \alpha n_1^b - \theta - p_1^s + \delta_s [V_{2I}^s(\theta) - V_{2E}^s(\theta)]. \quad (5)$$

Each continuation value is continuous and piecewise affine in θ , with slope either -1 or 0 . Consequently, the slope of $\Delta U_1^s(\theta)$ on any linear segment belongs to $\{-1 - \delta_s, -1, -1 + \delta_s\}$. Because $\delta_s \in (0, 1)$, all three slopes are strictly negative, so period-1 participation is characterized by a unique cutoff. Under the uniform distribution, sellers with $\theta \leq n_1^s$ participate in period 1, and the marginal early seller has type $\theta = n_1^s$.¹

Among period-1 sellers, the active incumbents in period 2 have types $\theta \leq n_{2I}^s$. Among sellers that waited, late entrants have types $\theta \in (n_1^s, n_1^s + n_{2E}^s]$.

For a participation profile that is interior on every active margin, the corresponding inverse-fee system is

$$p_1^s = \alpha n_1^b - n_1^s + \delta_s [V_{2I}^s(n_1^s) - V_{2E}^s(n_1^s)], \quad (6)$$

$$p_{2I}^s = \alpha n_2^b + c + x - n_{2I}^s, \quad (7)$$

$$p_{2E}^s = \alpha n_2^b - n_1^s - n_{2E}^s, \quad (8)$$

$$p_t^b = 1 + \beta n_t^s - n_t^b, \quad t = 1, 2. \quad (9)$$

The period-1 seller fee is pinned down whenever the early-participation margin is interior; at a boundary allocation, an implementing fee instead satisfies the relevant participation inequality. In period 2, the incumbent fee reflects acquired capability $c + x$ and the marginal incumbent type n_{2I}^s , whereas the entrant fee reflects the marginal entrant type $n_1^s + n_{2E}^s$. Hence, the fee gap combines acquired capability with endogenous selection across cohorts.

We study a pure-strategy perfect Bayesian equilibrium with monotone cutoff strategies. At the initial node, the platform chooses period-1 fees and support to maximize total discounted profit. After every public period-1 history, it chooses continuation fees to maximize period-2 profit given the inherited state and posterior beliefs. Agents participate optimally given expected participation masses, and these expectations coincide with the induced masses. Following a history summarized by an early-seller mass n_1^s , beliefs assign incumbent types according to the uniform prior truncated to $[0, n_1^s]$ and waiting-seller types according to the prior truncated to $(n_1^s, 1]$, consistent with the

¹Online Appendix 2.2 verifies that a fixed period-1 price vector and support level implement at most one early-seller mass across all relevant continuation activity sets.

monotone cutoff strategy. Indifferent agents participate; atomless types make this convention immaterial.

Assumption 1. *The cross-side network parameters α and β satisfy $\alpha + \beta < 1$.*

Together with $\delta_s \in (0, 1)$, $\delta_p \in (0, 1]$, and $\kappa > 0$, Assumption 1 makes the on-path reduced-profit branches used in the equilibrium characterization strictly concave. In the uniform baseline, it also implies $2\alpha\beta < 1$, which is sufficient for a unique period-2 fulfilled-expectations participation vector for every fixed history and price vector. The assumption places the entry-closure threshold strictly between zero and one and keeps all active participation margins interior on the positive-late-entry branch. Economically, it limits cross-side effects sufficiently to rule out within-period tipping.

3 Dynamic Seller Entry under Limited Commitment

3.1 The Option to Wait and the Early Acquisition Fee

We solve the game by backward induction. Fix early seller participation n_1^s and support x , and consider the continuation branch with full incumbent retention and positive late entry. On this branch, the period-2 allocation satisfies

$$n_2^b = \frac{2 + (\alpha + \beta)n_1^s}{4 - (\alpha + \beta)^2}, \quad (10)$$

$$n_{2I}^s = n_1^s, \quad (11)$$

$$n_{2E}^s = \frac{(\alpha + \beta) - [2 - (\alpha + \beta)^2]n_1^s}{4 - (\alpha + \beta)^2}. \quad (12)$$

Here, the period-2 first-order conditions determine buyer participation and late seller entry, while full incumbent retention implies $n_{2I}^s = n_1^s$.

The marginal early seller has type $\theta = n_1^s$. By entering early, this seller becomes the marginal incumbent and earns zero continuation rent. If the seller instead waits, it becomes the lowest-friction late entrant. Under the uniform type distribution, the distance between this seller's type and that of the marginal late entrant is n_{2E}^s , so its waiting rent equals n_{2E}^s . Substituting the period-2 fee expressions from (7) and (8) and imposing full incumbent retention gives

$$V_{2I}^s(n_1^s) = 0, \quad V_{2E}^s(n_1^s) = n_{2E}^s, \quad (13)$$

Using these continuation values, the marginal seller's indifference condition in (6) reduces to

$$p_1^s = \alpha n_1^b - n_1^s - \delta_s n_{2E}^s. \quad (14)$$

The first two terms are the seller's current cross-side benefit and operating friction. The final term deducts the discounted waiting rent that the platform must compensate to induce early participation. This waiting concession is the central dynamic wedge in the platform's early-acquisition problem. Substituting the implementing fee into (1) yields, for each early seller with $\theta_i \leq n_1^s$,

$$u_{i1}^{s*} = n_1^s - \theta_i + \delta_s n_{2E}^s \geq 0. \quad (15)$$

When late entry is absent, the waiting-rent term vanishes. A negative period-1 seller fee, $p_1^{s*} < 0$, represents a transfer from the platform to each early seller.

3.2 Equilibrium Regimes and the Waiting-Option Kink

The next theorem characterizes the global limited-commitment equilibrium.

Theorem 1 (Equilibrium allocation under limited commitment). *Under Assumption 1, the on-path equilibrium allocation is unique. Every seller who enters in period 1 remains active in period 2, so $n_{2I}^{s*} = n_1^{s*}$. The optimal support level is $x^* = \delta_p/\kappa$. The parameter space is partitioned into four regions.*

- (i) Positive late entry. If $\delta_p c + \frac{\delta_p^2}{2\kappa} < \frac{(1+\delta_p)(\alpha+\beta)}{2-(\alpha+\beta)^2} - \frac{\delta_s(\alpha+\beta)}{4-(\alpha+\beta)^2}$, then the early-seller mass is

$$n_1^{s*} = \frac{(\alpha + \beta)[4 - (\alpha + \beta)^2] + 2(\alpha + \beta)(\delta_p - \delta_s) + [4 - (\alpha + \beta)^2](2\delta_p c + \delta_p^2/\kappa)}{[4 - (\alpha + \beta)^2]^2 - 4\delta_s[2 - (\alpha + \beta)^2] + 4\delta_p[3 - (\alpha + \beta)^2]},$$

where the denominator is strictly positive under Assumption 1. Late entry is

$$n_{2E}^{s*} = \frac{(\alpha + \beta) - [2 - (\alpha + \beta)^2]n_1^{s*}}{4 - (\alpha + \beta)^2} > 0.$$

- (ii) Zero-late-entry plateau. If $\frac{(1+\delta_p)(\alpha+\beta)}{2-(\alpha+\beta)^2} - \frac{\delta_s(\alpha+\beta)}{4-(\alpha+\beta)^2} \leq \delta_p c + \frac{\delta_p^2}{2\kappa} \leq \frac{(1+\delta_p)(\alpha+\beta)}{2-(\alpha+\beta)^2}$, then

$$n_1^{s*} = \frac{\alpha + \beta}{2 - (\alpha + \beta)^2} \text{ and } n_{2E}^{s*} = 0.$$

(iii) Interior zero-late-entry regime. If $\frac{(1+\delta_p)(\alpha+\beta)}{2-(\alpha+\beta)^2} < \delta_p c + \frac{\delta_p^2}{2\kappa} < \frac{(1+\delta_p)[4-(\alpha+\beta)^2-(\alpha+\beta)]}{2}$, then

$$n_1^{s*} = \frac{(1+\delta_p)(\alpha+\beta) + 2\delta_p c + \delta_p^2/\kappa}{(1+\delta_p)[4-(\alpha+\beta)^2]} \text{ and } n_{2E}^{s*} = 0.$$

(iv) Seller saturation. If $\delta_p c + \frac{\delta_p^2}{2\kappa} \geq \frac{(1+\delta_p)[4-(\alpha+\beta)^2-(\alpha+\beta)]}{2}$, then

$$n_1^{s*} = n_{2I}^{s*} = 1 \text{ and } n_{2E}^{s*} = 0.$$

Holding support fixed, the platform's reduced profit is strictly decreasing in the inherited early-seller mass within each partial-retention region. An optimal period-1 policy therefore never recruits sellers whom the platform plans to exclude in period 2: every early seller remains active along the equilibrium path. Partial retention can nevertheless arise after an off-path history with a sufficiently large inherited seller base.² Full retention allows the platform to capture the return to support through the incumbent-seller fee. An increase in support raises every incumbent's willingness to pay one for one, allowing the platform to increase this fee without changing the continuation allocation. The net support-related contribution to profit is therefore $\delta_p x - \kappa x^2/2$ per early seller. It is maximized at $x^* = \delta_p/\kappa$, with value $\delta_p^2/(2\kappa)$. Adding the discounted return $\delta_p c$ from exogenous learning by doing yields the net capability contribution $\delta_p c + \delta_p^2/(2\kappa)$ per early seller. This same term appears in all three regime boundaries.

The plateau originates in the waiting concession embedded in the period-1 seller fee. When late entry is positive, equation (14) requires the platform to reduce the common fee by $\delta_s n_{2E}^s$. Because the reduction applies to all n_1^s early sellers, it lowers aggregate period-1 seller-fee revenue by $\delta_s n_1^s n_{2E}^s$. Late entry declines linearly with early participation and reaches zero at the entry-closure mass $\bar{n} = (\alpha + \beta)/[2 - (\alpha + \beta)^2]$. As early participation approaches $\bar{n}$ from below, the concession paid to each seller converges continuously to zero. An additional early seller nevertheless continues to reduce the concession paid to the entire existing cohort. Once late entry closes, it cannot decline further, and this cohort-wide marginal saving disappears. The reduced-profit function is therefore continuous but kinked at $\bar{n}$.³ Under the conditions for Region (ii), the kink is the unique maximizer

²See Online Appendix 2.1.

³Online Appendix 2.2 derives the reduced-profit functions on both sides of the entry-closure boundary and compares their one-sided derivatives.

and early participation remains fixed at $\bar{n}$. Below this parameter range, the optimum has positive late entry; above it, early recruitment resumes without late entry and eventually reaches seller saturation.

The next proposition completes the equilibrium characterization by reporting the remaining participation quantities and the associated fees in each region.

Proposition 1 (Equilibrium participation and prices). *In every region, period-1 participation and fees satisfy*

$$n_1^{b*} = \frac{1 + (\alpha + \beta)n_1^{s*}}{2}, \quad p_1^{b*} = \frac{1 + (\beta - \alpha)n_1^{s*}}{2}, \quad \text{and } p_1^{s*} = \alpha n_1^{b*} - n_1^{s*} - \delta_s n_{2E}^{s*}.$$

The continuation equilibrium is as follows.

(i) *In the positive-late-entry region, $n_{2E}^{s*} > 0$, and we have*

$$n_2^{b*} = \frac{2 + (\alpha + \beta)n_1^{s*}}{4 - (\alpha + \beta)^2}, \quad n_2^{s*} = n_1^{s*} + n_{2E}^{s*},$$

$$p_2^{b*} = 1 + \beta(n_1^{s*} + n_{2E}^{s*}) - n_2^{b*}, \quad p_{2I}^{s*} = \alpha n_2^{b*} + c + x^* - n_1^{s*}, \quad \text{and } p_{2E}^{s*} = \alpha n_2^{b*} - n_1^{s*} - n_{2E}^{s*}.$$

(ii) *In the zero-late-entry plateau, we have*

$$n_1^{b*} = n_2^{b*} = \frac{1}{2 - (\alpha + \beta)^2}, \quad n_1^{s*} = n_2^{s*} = \frac{\alpha + \beta}{2 - (\alpha + \beta)^2},$$

$$p_1^{b*} = p_2^{b*} = \frac{1}{2} \left[1 + \frac{(\beta - \alpha)(\alpha + \beta)}{2 - (\alpha + \beta)^2} \right],$$

$$p_1^{s*} = -\frac{\beta}{2 - (\alpha + \beta)^2}, \quad p_{2I}^{s*} = c + \frac{\delta_p}{\kappa} - \frac{\beta}{2 - (\alpha + \beta)^2}.$$

Moreover, $n_{2E}^{s} = 0$, and any $p_{2E}^{s*} \geq -\beta/[2 - (\alpha + \beta)^2]$ supports this allocation.*

(iii) *In the interior zero-late-entry region, we have*

$$n_1^{b*} = n_2^{b*} = \frac{1 + (\alpha + \beta)n_1^{s*}}{2}, \quad n_2^{s*} = n_1^{s*},$$

$$p_1^{b*} = p_2^{b*} = \frac{1 + (\beta - \alpha)n_1^{s*}}{2}, \quad p_1^{s*} = \frac{\alpha[1 + (\alpha + \beta)n_1^{s*}]}{2} - n_1^{s*}, \quad \text{and } p_{2I}^{s*} = p_1^{s*} + c + \frac{\delta_p}{\kappa}.$$

Moreover, $n_{2E}^{s} = 0$, and any $p_{2E}^{s*} \geq p_1^{s*}$ supports this allocation.*

(iv) *Under seller saturation, we have*

$$n_1^{b*} = n_2^{b*} = \frac{1 + \alpha + \beta}{2}, \quad n_1^{s*} = n_2^{s*} = 1,$$

$$p_1^{b*} = p_2^{b*} = \frac{1 + \beta - \alpha}{2}, \quad p_1^{s*} = \frac{\alpha(1 + \alpha + \beta)}{2} - 1, \quad \text{and } p_{2I}^{s*} = p_1^{s*} + c + \frac{\delta_p}{\kappa}.$$

Moreover, $n_{2E}^{s} = 0$, and any $p_{2E}^{s*} \geq p_1^{s*}$ supports this allocation.*

The proposition shows how waiting rents, acquired capability, and selection across seller cohorts shape platform fees. In Region (i), the period-1 seller fee equals the marginal seller's current willingness to pay, $\alpha n_1^{b*} - n_1^{s*}$, less the discounted rent it would earn by waiting until period 2, $\delta_s n_{2E}^{s*}$. The period-2 seller-fee gap decomposes into acquired capability and endogenous selection:

$$p_{2I}^{s*} - p_{2E}^{s*} = c + \delta_p/\kappa + n_{2E}^{s*}. \quad (16)$$

The term $c + \delta_p/\kappa$ is the capability acquired through early participation. The term n_{2E}^{s*} is the type gap between the marginal incumbent, whose type is n_1^{s*} , and the marginal late entrant, whose type is $n_1^{s*} + n_{2E}^{s*}$.

In Regions (ii)–(iv), $n_{2E}^{s*} = 0$, so the waiting concession in the period-1 seller fee vanishes. Because the seller mass is unchanged and the buyer environment is otherwise stationary, buyer participation and the buyer fee are identical across periods. The incumbent fee exceeds the period-1 seller fee by exactly $c + \delta_p/\kappa$, allowing the platform to capture acquired capability in period 2.

In Region (ii), participation depends on the aggregate cross-side effect $\alpha + \beta$, whereas period-1 fees also depend on the relative magnitudes of α and β . Participation is locally invariant to c , δ_p , δ_s , and κ . In Region (iii), by contrast, n_1^{s*} varies with $\alpha + \beta$, c , δ_p , and κ , but not with δ_s . Buyer participation and fees on active margins vary with n_1^{s*} , while the incumbent fee also contains the direct capability component $c + \delta_p/\kappa$. Region (iv) begins when $n_1^s = 1$ and the seller-population constraint binds.

Finally, when $n_{2E}^{s*} = 0$, any entrant fee at or above the entry-detering threshold (the choke fee) sustains zero late entry. The fee is posted on the equilibrium path but is not paid, and its exact value has no effect on the allocation or platform profit. It is therefore set-valued in Regions (ii)–(iv).

Corollary 1 (Platform patience and capability thresholds). *Holding κ fixed, write $\delta_p = \kappa x^*$ and express the three capability thresholds in Theorem 1 as functions of $x^* \in (0, 1/\kappa]$:*

$$\begin{aligned} c_1(x^*) &= \frac{\alpha + \beta}{2 - (\alpha + \beta)^2} + \frac{1}{\kappa x^*} \left[\frac{\alpha + \beta}{2 - (\alpha + \beta)^2} - \frac{\delta_s(\alpha + \beta)}{4 - (\alpha + \beta)^2} \right] - \frac{x^*}{2}, \\ c_2(x^*) &= \frac{\alpha + \beta}{2 - (\alpha + \beta)^2} + \frac{\alpha + \beta}{\kappa x^* [2 - (\alpha + \beta)^2]} - \frac{x^*}{2}, \\ c_3(x^*) &= \frac{4 - (\alpha + \beta)^2 - (\alpha + \beta)}{2} + \frac{4 - (\alpha + \beta)^2 - (\alpha + \beta)}{2\kappa x^*} - \frac{x^*}{2}. \end{aligned}$$

The thresholds $c_1(x^*)$, $c_2(x^*)$, and $c_3(x^*)$ separate Regions (i) and (ii), Regions (ii) and (iii), and Regions (iii) and (iv), respectively. Each threshold is strictly decreasing and strictly convex in x^* .

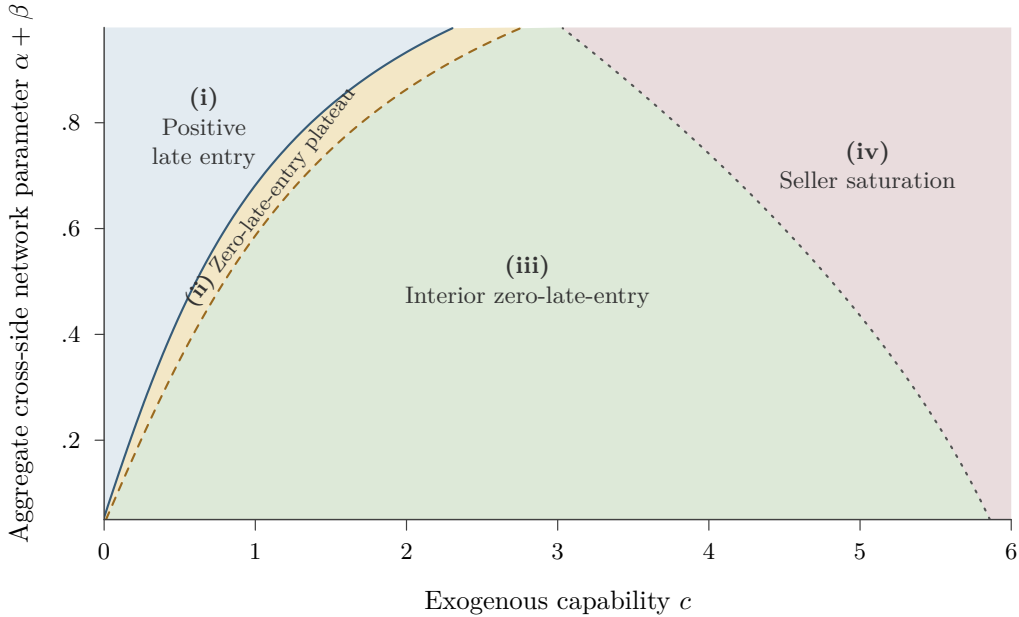


Figure 3: Capability, Cross-Side Effects, and Equilibrium Regimes

Notes. The figure fixes $x^* = 0.125$, $\delta_s = 0.70$, and $\kappa = 4$, implying $\delta_p = 0.50$. The horizontal axis reports exogenous capability c , and the vertical axis reports $\alpha + \beta \in (0, 1)$. Regions (i)–(iv) correspond to the four cases in Theorem 1. The solid, dashed, and dotted curves denote the capability thresholds c_1 , c_2 , and c_3 , respectively, and mark the successive transitions from positive late entry to the zero-late-entry plateau, then to the interior zero-late-entry region, and finally to seller saturation.

Holding x^* , δ_s , and κ fixed, Figure 3 maps the equilibrium regions in $(c, \alpha + \beta)$ space. Corollary 1 complements this map by showing how the capability thresholds vary with x^* .⁴ For any fixed $\alpha + \beta$, a higher c increases the return to recruiting sellers in period 1 because only early sellers acquire capability. If $c_1 > 0$, increasing c produces the full sequence from positive late entry through the

⁴Online Appendix 2.5 presents two additional regime maps. With κ fixed, varying $x^* = \delta_p/\kappa$ changes both the weight on continuation profit and the optimal support level; it is therefore not a pure exogenous support shock.

two zero-late-entry regions to seller saturation. A larger aggregate cross-side effect shifts the first two thresholds upward. The entry-closure mass $\bar{n} = (\alpha + \beta)/[2 - (\alpha + \beta)^2]$ rises, so more early sellers are required to eliminate late entry and subsequently to move beyond the plateau. The gap $c_2 - c_1$ also widens because the waiting concession increases with $\alpha + \beta$ and affects c_1 but not c_2 . The saturation threshold moves in the opposite direction. A larger $\alpha + \beta$ raises the cross-side revenue generated by a larger seller base, reducing the capability required for n_1^{s*} to reach one. Thus, stronger cross-side effects delay both the disappearance of late entry and departure from the plateau, but hasten seller saturation.

3.3 Early-Seller Subsidization

Proposition 2 (Early-seller subsidization). *Let c^S denote the unique zero of the affine early-seller fee implied by the Region (i) equilibrium formulas.⁵ Thus, c^S is characterized by*

$$p_1^{s*}(c^S) = \alpha n_1^{b*}(c^S) - n_1^{s*}(c^S) - \delta_s n_{2E}^{s*}(c^S) = 0.$$

If Region (i) is nonempty, p_1^{s} is strictly decreasing in c , and within that region,*

$$p_1^{s*} < 0 \text{ if and only if } c > c^S.$$

In Regions (ii)–(iv), the early-seller fee is strictly negative.

The proposition shows that the platform can charge a positive early-seller fee only while late entry remains active; within that region, it does so if and only if $c < c^S$. If Region (i) is empty, the equilibrium at $c = 0$ lies on the plateau or in a later regime, and early sellers are subsidized for every feasible c . The marginal early seller receives the current cross-side benefit αn_1^{b*} , but early participation also requires the platform to offset the seller's operating friction and, while late entry remains active, the discounted rent forgone by not waiting. The fee is negative when these acquisition requirements exceed the seller's current willingness to pay. The platform is willing to pay this subsidy because an installed seller generates current seller-to-buyer revenue and future incumbent revenue, including the capability surplus that the platform can capitalize.

⁵Online Appendix 2.7 provides the closed-form expression for c^S in primitive parameters.

Capability lowers the fee through different mechanisms on the two interior branches. In Region (i), a higher c expands the early cohort and requires the platform to recruit a higher-friction marginal seller. The induced decline in late entry reduces the waiting concession, but the first effect dominates, so p_1^{s*} falls. In Region (iii), no waiting concession remains, and capability lowers the fee solely by extending recruitment to higher-friction sellers. Within Region (i), greater seller patience also lowers the fee by raising the value of waiting. On either interior branch, a larger κ reduces support, weakens the return to early acquisition, and raises the fee.

The entry-closure boundary isolates the seller-to-buyer component of the subsidy. At this boundary, the waiting concession is zero, yet the Region (ii) fee, $p_1^{s*} = -\beta/[2 - (\alpha + \beta)^2]$, remains strictly negative. The platform subsidizes early sellers because their presence generates buyer-side revenue. Beyond the plateau, stronger capability returns induce the platform to recruit sellers with still greater operating friction, deepening the subsidy even though no seller waits in equilibrium.⁶

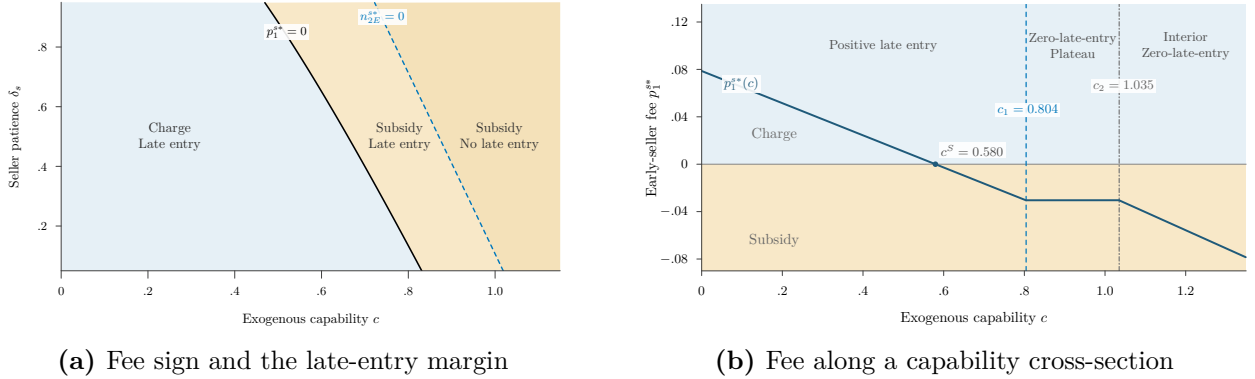


Figure 4: Early-Seller Subsidization and the Late-Entry Margin

Notes. Both panels fix $\alpha = 0.55$, $\beta = 0.05$, $\delta_p = 0.50$, and $\kappa = 4$. Panel **a** varies seller patience δ_s and exogenous capability c . Its solid curve is the analytical fee-zero boundary $c^S(\delta_s)$, and its dashed curve marks the disappearance of late entry. The fee is negative to the right of the solid curve. Panel **b** fixes $\delta_s = 0.70$. The fee crosses zero at $c^S = 0.580$, late entry reaches zero at $c_1 = 0.804$, and the plateau ends at $c_2 = 1.035$. Seller saturation occurs at $c_3 = 4.498$ and lies outside the plotted range. Types are independently uniform on $[0, 1]$, and the calibration is illustrative rather than estimated. A negative fee is a monetary transfer from the platform to an early seller, not a resource cost.

Figure 4 illustrates the subsidy mechanism from two perspectives. Panel (a) maps where the platform charges or subsidizes early sellers and locates the fee-zero boundary relative to the disappearance of late entry. Panel (b) traces the fee along a capability cross-section. The platform first switches from charging to subsidizing early sellers; late entry then disappears; the fee remains

⁶These results rely on unrestricted transfers. Imposing a nonnegative-fee constraint would change both the implemented allocation and the regime boundaries.

constant on the zero-late-entry plateau; and the subsidy deepens when early acquisition resumes in the interior zero-late-entry region.

3.4 Capability, Seller Patience, and Entry Timing

We next examine how capability and seller patience affect the timing and scale of seller participation.

Proposition 3 (Capability, seller patience, and entry timing). *For parameter changes that keep the equilibrium within a given region, the following comparative statics hold.*

1. In Region (i), greater capability shifts seller participation toward period 1 while expanding the period-2 seller base:

$$\frac{\partial n_1^{s*}}{\partial c} > 0, \quad \frac{\partial n_{2E}^{s*}}{\partial c} < 0, \quad \text{and} \quad \frac{\partial n_2^{s*}}{\partial c} > 0.$$

Buyer participation increases with capability in both periods, but the period-1 response is stronger. Consequently, period-2 buyer participation remains higher, while the intertemporal increase narrows:

$$n_2^{b*} > n_1^{b*}, \quad \frac{\partial n_1^{b*}}{\partial c} > \frac{\partial n_2^{b*}}{\partial c} > 0, \quad \text{and} \quad \frac{\partial (n_2^{b*} - n_1^{b*})}{\partial c} < 0.$$

Seller patience instead has an installed-base-dependent effect:

$$\text{sgn} \left(\frac{\partial n_1^{s*}}{\partial \delta_s} \right) = \text{sgn} \left(\frac{\partial n_2^{s*}}{\partial \delta_s} \right) = -\text{sgn} \left(\frac{\partial n_{2E}^{s*}}{\partial \delta_s} \right) = \text{sgn} \left(n_1^{s*} - \frac{\bar{n}}{2} \right).$$

Here, $\bar{n} = (\alpha + \beta) / [2 - (\alpha + \beta)^2]$ is the early-seller mass at which late entry disappears. Thus, if $n_1^{s*} < \bar{n}/2$, greater seller patience reduces period-1 and period-2 seller participation while increasing late entry. If $n_1^{s*} > \bar{n}/2$, all three responses reverse.

2. In Regions (ii)–(iv), seller patience has no local effect on the allocation.
3. In Regions (ii) and (iv), capability has no local effect on the allocation.
4. In Region (iii), however, capability raises both the common seller mass and the common buyer mass:

$$\frac{\partial n_1^{s*}}{\partial c} = \frac{\partial n_2^{s*}}{\partial c} > 0 \quad \text{and} \quad \frac{\partial n_1^{b*}}{\partial c} = \frac{\partial n_2^{b*}}{\partial c} > 0.$$

In Region (i), late entry is positive and both intertemporal margins are active. Capability affects both the timing and the scale of market development. Only period-1 sellers acquire $c + x^*$, so a higher c raises the continuation return from early recruitment and expands the initial seller cohort. Late entry declines by less than period-1 seller participation rises, and the period-2 seller base therefore expands. The larger seller bases, in turn, raise buyer participation in both periods through cross-side effects. Because the period-1 buyer response is stronger, the interperiod gap $n_2^{b*} - n_1^{b*}$ narrows.

Seller patience operates through two opposing forces. A higher δ_s increases the discounted rent from waiting and hence the concession required to induce period-1 participation. This direct acquisition-cost effect tends to reduce the initial seller cohort. Recruiting more sellers early, however, reduces late entry and the waiting concession embedded in the common period-1 fee. Because this concession applies to every early seller, the resulting cohort-wide saving becomes more important as the installed base expands. The direct cost effect dominates when $n_1^{s*} < \bar{n}/2$; the cohort-wide concession-saving effect dominates when $n_1^{s*} > \bar{n}/2$.

The remaining regions have simpler comparative statics. In Region (ii), the kink pins period-1 seller participation at $\bar{n}$ and eliminates late entry, so the allocation is locally invariant to both capability and seller patience. In Region (iii), waiting has no equilibrium value and seller patience has no local allocation effect. Greater capability raises the seller mass by the same amount in both periods and likewise generates identical buyer responses across periods. In Region (iv), the seller-population constraint binds, leaving the allocation locally invariant to both capability and seller patience.

Corollary 2 (Capability and seller-fee polarization). *In Region (i), capability lowers both the period-1 seller fee and the late-entrant fee but raises the incumbent fee:*

$$\frac{\partial p_1^{s*}}{\partial c} < 0, \quad \frac{\partial p_{2E}^{s*}}{\partial c} < 0, \quad \text{and} \quad \frac{\partial p_{2I}^{s*}}{\partial c} > 0.$$

The incumbent fee exceeds the late-entrant fee, and the gap increases by less than one for one with capability:

$$p_{2I}^{s*} - p_{2E}^{s*} > 0 \quad \text{and} \quad 0 < \frac{\partial(p_{2I}^{s*} - p_{2E}^{s*})}{\partial c} < 1.$$

Corollary 2 highlights that capability affects seller fees differently across dates and cohorts. As c rises, the platform expands the period-1 seller cohort, extending recruitment to sellers with greater operating friction. For the marginal early seller, this increase in friction dominates both the rise in buyer-side benefits and the decline in the waiting concession, so p_1^{s*} falls. The late-entrant fee also falls: although period-2 buyer participation rises, the increase in the marginal entrant's operating friction is larger. By contrast, the incumbent fee rises because the direct capitalization of capability dominates the increase in the marginal incumbent's operating friction.

These responses widen the incumbent–entrant fee gap. Equation (16) decomposes the gap into the direct capability component $c + \delta_p/\kappa$ and the endogenous selection component n_{2E}^{s*} . A higher c raises the first component one for one but reduces the second by shifting entry toward period 1. Selection therefore attenuates, but does not overturn, the direct capability effect. This comparison applies only in Region (i), where both continuation seller fees are active and uniquely determined. Once late entry disappears, the unused entrant fee is set-valued, so the incumbent–entrant fee gap has no unique equilibrium value.

4 Full Commitment, Market Development, and Social Welfare

The limited-commitment equilibrium exhibits a time-inconsistency problem. Once the period-1 seller base is installed, the platform reoptimizes the late-entrant fee without internalizing how the anticipated continuation terms affect sellers' ex ante value of waiting and hence the cost of early recruitment. To isolate this force, we allow the platform to commit, before period-1 participation, to the complete schedule of period-2 buyer, incumbent-seller, and entrant-seller fees following every public period-1 history. Sellers remain free to choose whether to participate in period 2. Thus, both regimes permit history-contingent pricing; they differ only in whether the continuation schedule is chosen ex ante or sequentially. Throughout this section, the platform and sellers share the discount factor $\delta_p = \delta_s \equiv \delta \in (0, 1)$, and superscripts C and L denote full and limited commitment, respectively.⁷

⁷Common discounting makes the timing of transfers neutral in present value and isolates the allocative effect of commitment. If $\delta_s > \delta_p$, shifting a payment from period 2 to period 1 while preserving seller utility raises platform present-value revenue and, absent liquidity or fee constraints, makes the problem unbounded. If $\delta_s < \delta_p$, the reverse shift is bounded by the seller's period-2 participation constraint, but it introduces an additional payment-timing and retention margin.

4.1 The Full-Commitment Model

Let n denote the mass of sellers recruited in period 1, $r \leq n$ the mass retained in period 2, and e the mass entering in period 2. The period-2 seller mass is $m \equiv r + e$. For any seller mass z , let

$$\phi(z) \equiv \frac{[1 + (\alpha + \beta)z]^2}{4} - z^2$$

denote the maximum one-period access-fee revenue generated by a seller mass z , after optimizing buyer participation and imposing the marginal seller's participation condition.⁸

The full-commitment problem. After optimizing buyer participation and substituting the sellers' lifetime participation conditions, the platform solves

$$\max_{\substack{0 \leq r \leq n \leq 1 \\ 0 \leq e \leq 1-n \\ x \geq 0}} \left\{ \phi(n) + \delta\phi(r+e) + \delta(c+x)r - 2\delta e(n-r) - \frac{\kappa x^2}{2}n \right\}.$$

The terms $\phi(n)$ and $\delta\phi(r+e)$ represent optimized access-fee revenue in periods 1 and 2, respectively. Retained early sellers generate the capability return $\delta(c+x)r$, whereas every early seller receives support at real cost $\kappa x^2 n/2$. The final term is the signed net effect of two intertemporal revenue adjustments. First, inducing early entry lowers period-1 seller-fee revenue by δen .⁹ Second, pricing retained incumbents and late entrants at their respective marginal types changes period-2 seller-fee revenue by $\delta e(2r-n)$.¹⁰ Hence their signed sum is $-\delta en + \delta e(2r-n) = -2\delta e(n-r)$; it vanishes under full retention, $r = n$. For $n > 0$, the conditional support choice is $x = \delta r/(\kappa n)$; if $n = 0$, support is payoff irrelevant and is normalized to zero. Under full retention, $x = \delta/\kappa$. Define the resulting discounted net capability value per early seller as $\mathcal{V} \equiv \delta c + \delta^2/(2\kappa)$.

⁸For a contiguous active seller set of mass z and buyer mass b , the inverse participation conditions are $p^b = 1 + \beta z - b$ and $p^s = \alpha b - z$. One-period access-fee revenue is therefore

$$p^b b + p^s z = [1 + (\alpha + \beta)z]b - b^2 - z^2.$$

Maximizing over b gives $b = [1 + (\alpha + \beta)z]/2$, and substitution yields $\phi(z)$. The term $-z^2$ is the revenue consequence of setting the seller fee to attract the marginal type $\theta = z$; it is not the aggregate real operating cost, which equals $\int_0^z \theta d\theta = z^2/2$. Capability returns, waiting compensation, and support costs enter the dynamic objective separately.

⁹The marginal early seller has type n . At the entrant fee $p_{2E}^s = \alpha n_2^b - n - e$, it would earn continuation rent e by waiting. Its period-1 participation condition therefore requires $p_1^s = \alpha n_1^b - n - \delta e$. Because this common fee applies to all n early sellers, period-1 seller-fee revenue is lower by δen .

¹⁰Excluding the capability return, cohort-specific seller-fee revenue is $r(\alpha n_2^b - r) + e(\alpha n_2^b - n - e)$. The seller-fee revenue already included in $\phi(r+e)$ is $\alpha n_2^b(r+e) - (r+e)^2$. Their difference is $e(2r-n)$. Under full retention, $r = n$, this difference is en : the resulting discounted continuation revenue δen exactly offsets the period-1 waiting concession under common discounting.

Proposition 4 (Complete front-loading of seller participation). *Under Assumption 1 and common discounting, the participation allocation under full commitment is unique. All participating sellers enter in period 1 and remain active in period 2. We have:*

$$\begin{aligned} n_1^{s,C} = n_{2I}^{s,C} = n_2^{s,C} &= \min \left\{ 1, \frac{(1+\delta)(\alpha+\beta) + 2\mathcal{V}}{(1+\delta)[4 - (\alpha+\beta)^2]} \right\}, \\ n_{2E}^{s,C} &= 0, \quad n_1^{b,C} = n_2^{b,C} = \frac{1 + (\alpha+\beta)n_1^{s,C}}{2}. \end{aligned}$$

An early seller who is not retained has received costly support but generates no continuation capability revenue. If late entry is positive, replacing a late entrant with such an early seller raises profit: the early seller has a lower operating-friction type and acquired capability, while the total period-2 seller mass is unchanged. If late entry is zero, partial retention is dominated either by retaining more early sellers or by recruiting fewer sellers initially. Hence, $r = n$ and $m = n + e \geq n$. After optimizing support, the problem reduces to

$$\max_{0 \leq n \leq m \leq 1} \{ \phi(n) + \delta \phi(m) + \mathcal{V}n \}.$$

The reduced form does not make early acquisition costless: $\phi(n)$ incorporates the revenue loss from extending participation to higher-friction sellers, and support remains a real resource cost. Common discounting eliminates only the present-value transfer wedge associated with waiting. Equivalently, under full retention and $e = m - n$, profit before optimizing support can be written as

$$\underbrace{\phi(n) - \delta ne - \frac{\kappa x^2}{2}n}_{\text{period 1}} + \underbrace{\delta [\phi(m) + (c+x)n + ne]}_{\text{period 2}}.$$

The term $-\delta ne$ is the period-1 revenue loss required to neutralize waiting, whereas $+\delta ne$ is the additional continuation revenue collected from the installed cohort. They cancel because sellers and the platform use the same discount factor. The support expenditure $\kappa x^2 n/2$ remains because it uses real resources.

Absent the constraint $n \leq m$, the early-seller target exceeds the period-2 seller target because $\mathcal{V} > 0$. Strict concavity therefore makes $n \leq m$ bind, implying $m = n$. By choosing the entrant fee ex ante, the platform internalizes how continuation pricing affects the value of waiting and recruits

in period 1 every seller it intends to serve in period 2. Late entry and subsequent buyer growth consequently disappear. The allocation can be implemented by committing to an entrant fee at or above the entry-detering threshold, $p_{2E}^{s,C} \geq \alpha n_2^{b,C} - n_1^{s,C}$.

The full-commitment seller mass follows the same formula as the interior zero-late-entry allocation, Region (iii), under limited commitment. Full commitment extends this allocation rule to parameter values at which it is not sequentially implementable and thereby removes both the positive-late-entry branch and the plateau. Participation and support are unique, but seller fees need not be: the unused entrant fee is set-valued, and common discounting makes the division of incumbent payments across periods neutral in present value. Online Appendix 3.1 provides supporting fees and the formal proof.

4.2 Market Development and Platform Profit

To compare the two commitment regimes, define three capability-value thresholds:

$$\mathcal{V}_\times = \frac{2(1+\delta)(\alpha+\beta)}{[4-(\alpha+\beta)^2]^2-4}, \quad \mathcal{V}_1 = \frac{(1+\delta)(\alpha+\beta)}{2-(\alpha+\beta)^2} - \frac{\delta(\alpha+\beta)}{4-(\alpha+\beta)^2}, \quad \mathcal{V}_2 = \frac{(1+\delta)(\alpha+\beta)}{2-(\alpha+\beta)^2}.$$

They satisfy $0 < \mathcal{V}_\times < \mathcal{V}_1 < \mathcal{V}_2$. Let Π^C and Π^L denote optimized platform profit under full and limited commitment.

Proposition 5 (Commitment, market development, and platform profit). *Under Assumption 1 and common discounting, limited commitment generates late entry if and only if $\mathcal{V} < \mathcal{V}_1$, whereas full commitment generates no late entry:*

$$n_{2E}^{s,L} > 0 \iff \mathcal{V} < \mathcal{V}_1, \quad n_{2E}^{s,C} = 0.$$

Early seller participation satisfies

$$\begin{aligned} n_1^{s,C} > n_1^{s,L} &\iff \mathcal{V} < \mathcal{V}_\times, \\ n_1^{s,C} = n_1^{s,L} &\iff \mathcal{V} = \mathcal{V}_\times \text{ or } \mathcal{V} \geq \mathcal{V}_2, \\ n_1^{s,C} < n_1^{s,L} &\iff \mathcal{V}_\times < \mathcal{V} < \mathcal{V}_2. \end{aligned}$$

Period-1 buyer participation has the same ranking. Limited commitment yields strictly greater period-2 participation on both sides if and only if $\mathcal{V} < \mathcal{V}_2$:

$$n_2^{s,L} > n_2^{s,C} \text{ and } n_2^{b,L} > n_2^{b,C} \iff \mathcal{V} < \mathcal{V}_2.$$

Platform profit satisfies

$$\Pi^C > \Pi^L \iff \mathcal{V} < \mathcal{V}_2, \quad \Pi^C = \Pi^L \iff \mathcal{V} \geq \mathcal{V}_2.$$

The thresholds separate three comparisons. At $\mathcal{V}_x$, the regimes recruit the same early seller mass even though limited commitment still permits late entry. At $\mathcal{V}_1$, late entry under limited commitment disappears and the kink pins early recruitment at the entry-closure mass. At $\mathcal{V}_2$, the full-commitment allocation reaches that mass and the two regimes coincide. Because buyer participation increases with the contemporaneous seller mass, limited commitment generates period-2 buyer growth if and only if $\mathcal{V} < \mathcal{V}_1$; full commitment generates none.

At low capability values, commitment removes the marginal seller's return to waiting and lowers the cost of early acquisition, so full commitment recruits more sellers initially. Limited commitment nevertheless expands further after the initial seller base is sunk and therefore serves more sellers and buyers in period 2. At intermediate values, limited commitment recruits more sellers early because doing so shrinks the residual entrant pool and reduces the waiting concession paid to the entire early cohort. Once late entry closes, the kink holds limited-commitment recruitment at the closure mass while full-commitment recruitment approaches that mass from below. Zero late entry is therefore insufficient for allocation equivalence.

Commitment thus creates a profit–market-development tradeoff. The limited-commitment equilibrium plan is feasible under full commitment, so commitment cannot reduce platform profit. The gain is strict whenever the unique participation allocations differ. It arises from removing the sequential-credibility constraint, not from enlarging the continuation market: whenever commitment changes the allocation, it raises platform profit but reduces period-2 participation on both sides.

4.3 Social Welfare

We evaluate discounted social welfare at the two equilibrium allocations rather than solve a planner's first-best problem. Social welfare is the sum of buyer utility, seller utility, and platform profit, using δ as the social discount factor. Access fees reduce participants' utility by exactly the amount they add to platform profit, so they cancel as transfers; seller operating frictions and platform support remain real resource costs.

Consider a period with seller mass z and buyer mass b . The cutoff strategies imply that participating buyers have types $v \in [1-b, 1]$ and participating sellers have types $\theta \in [0, z]$. Excluding acquired capability and support costs, aggregate surplus is

$$\Omega(b, z) \equiv \int_{1-b}^1 (v + \beta z) dv + \int_0^z (\alpha b - \theta) d\theta = b - \frac{b^2}{2} + (\alpha + \beta)bz - \frac{z^2}{2}.$$

The first integral combines buyers' stand-alone values and cross-side benefits; the second combines sellers' cross-side benefits and operating frictions. In particular, $\int_0^z \theta d\theta = z^2/2$ is the aggregate real operating cost. It differs from the term $-z^2$ in $\phi(z)$, which is a platform-revenue consequence of lowering the common seller fee to attract the marginal type. At the platform equilibrium, buyer participation conditional on seller mass z is $b(z) = [1 + (\alpha + \beta)z]/2$. Substitution gives one-period equilibrium welfare, before capability and support, as

$$\omega(z) = \Omega(b(z), z) = \frac{3 + 6(\alpha + \beta)z + [3(\alpha + \beta)^2 - 4]z^2}{8}.$$

Now let n sellers enter in period 1. Full retention in both regimes implies that all n remain active in period 2, when the total seller mass is $m \geq n$. The period-1 market generates $\omega(n)$ and incurs the real support cost $\kappa x^2 n/2$. The period-2 market generates $\omega(m)$. Because each of the n retained early sellers receives the same capability increment $c + x$, capability adds $(c + x)n$ to period-2 surplus. Hence, before optimizing support, discounted welfare is

$$W(n, m; x) = \omega(n) - \frac{\kappa x^2}{2}n + \delta [\omega(m) + (c + x)n].$$

Holding the participation masses (n, m) fixed, $\partial W/\partial x = n(\delta - \kappa x)$, so the conditionally welfare-maximizing support level is $x = \delta/\kappa$. Both commitment regimes choose this level because, under full retention, the platform can capitalize the marginal capability increment into the incumbent fee one for one. Substituting it and using $\mathcal{V} \equiv \delta c + \delta^2/(2\kappa)$ yields

$$W(n, m) = \omega(n) + \delta\omega(m) + \mathcal{V}n.$$

Let W^C and W^L denote this welfare measure evaluated at the full- and limited-commitment equilibrium allocations, respectively.

Proposition 6 (Commitment and social welfare). *Under Assumption 1 and common discounting,*

$$W^L > W^C \iff \mathcal{V} < \mathcal{V}_2, \quad W^L = W^C \iff \mathcal{V} \geq \mathcal{V}_2.$$

The welfare ranking follows from the exact surplus differences, not from assuming that welfare is monotone in seller participation: additional sellers create cross-side benefits but also have higher operating frictions. When $\mathcal{V} < \mathcal{V}_\times$, full commitment recruits more sellers initially, but limited commitment's larger period-2 market produces a strictly larger discounted surplus overall. At $\mathcal{V} = \mathcal{V}_\times$, early participation is identical and the continuation market remains strictly larger under limited commitment. For $\mathcal{V}_\times < \mathcal{V} < \mathcal{V}_2$, limited commitment generates greater participation in both periods; the exact welfare difference remains positive after subtracting the additional sellers' operating frictions. At and above $\mathcal{V}_2$, the allocations coincide, and welfare is therefore equal.

Support intensity does not drive the ranking: holding participation fixed, both regimes choose the welfare-maximizing level $x = \delta/\kappa$. The welfare difference instead reflects the timing and scale of participation, the associated cross-side benefits, and seller operating frictions. Thus, whenever commitment changes the allocation, its private profit gain is accompanied by a smaller period-2 market and lower social welfare. This conclusion depends on the benchmark's common discounting, unrestricted transfers, and absence of external development benefits. Figure 5 illustrates the comparison.

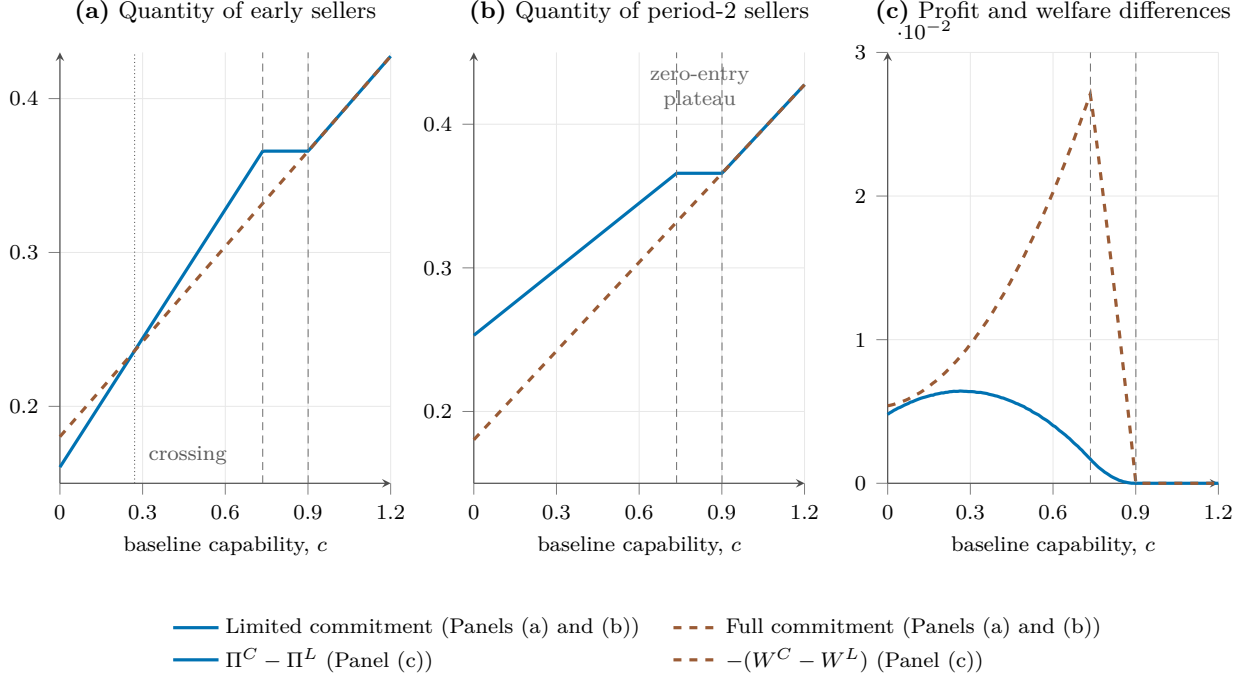


Figure 5: Commitment, Market Development, and Social Welfare

Notes. Types are independently uniform on $[0, 1]$. Parameters are $\alpha = 0.35$, $\beta = 0.25$, $\delta = 0.60$, and $\kappa = 4$; c varies on the horizontal axis. Solid blue and dashed brown lines denote limited and full commitment, respectively, in Panels (a) and (b). The vertical lines at $c = 0.271$, $c = 0.736$, and $c = 0.901$ correspond to $\mathcal{V}_\times$, $\mathcal{V}_1$, and $\mathcal{V}_2$. Panel (c) reports the platform-profit gain from commitment and the social-welfare advantage of limited commitment.

5 Empirical and Managerial Implications

The model yields region-specific predictions for seller entry, cohort pricing, and market development. An empirical implementation should define comparable launch and continuation windows and construct cohort-level panels that distinguish period-1 active sellers, their continued activity, sellers first active in period 2, and the resulting period-2 seller base. Active participation, rather than account registration, is the relevant outcome because only active agents generate cross-side benefits and incur operating costs. Comparisons should also distinguish movements within a region from transitions across region boundaries.

Measurement and identification. Monetary concessions and capability support should be measured separately. A seller fee measures the platform’s net monetary receipt from the seller, after commissions, rebates, credits, or direct grants. Support instead consists of real inputs such as training, integration software, logistics assistance, or account-management effort. Combining these

objects into a single seller-acquisition-cost measure would obscure the model’s central distinction between transfers that implement participation and expenditures that create productive capability.

The exogenous capability component c is a monetary-equivalent increment to an incumbent seller’s continuation payoff attributable to early active participation after measured support is held fixed. Changes in fulfillment speed, conversion, product quality, capacity, or operating cost are potential empirical proxies rather than direct measures in the model’s units. Estimating c requires separating capability formation from selection into early entry and from the direct effect of support x . Support should be measured using real inputs or expenditure per early seller, while its cost curvature κ should be determined by the platform’s resource-cost schedule. Within the benchmark, $x^* = \delta_p/\kappa$, so support measured in model units identifies this ratio rather than its two components. Region membership depends on c and κ through $\delta_p c + \delta_p^2/(2\kappa)$, while δ_p also enters the regime thresholds directly. Therefore, observed support and regime classification do not separately identify c , δ_p , and κ . The two cross-side effects require distinct variation in buyer exposure to active sellers and seller exposure to buyers. Variables such as liquidity, working-capital access, planning horizons, or managerial incentives may proxy for seller or platform patience, but a causal interpretation requires an exclusion restriction ruling out direct effects on productivity and demand.

Regime classification. Under limited commitment, positive late entry is diagnostic of Region (i) and an active waiting margin. Zero late entry does not distinguish Regions (ii)–(iv), and it also arises under full commitment. Within limited commitment, these cases can be separated by combining coverage measures with responses to plausibly exogenous changes in baseline capability. On the plateau, early and period-2 participation are locally invariant to capability; in the interior zero-late-entry region, seller participation rises by the same amount in both periods and buyer participation does likewise; under saturation, the seller-population constraint prevents further expansion. Thus, observing no late entry is insufficient to infer either commitment or a locally unresponsive market. The model also predicts full incumbent retention on the equilibrium path. Material attrition should be interpreted as evidence of an off-path inherited seller base or of forces outside the benchmark rather than reclassified as late entry.

Within-region predictions. When late entry is positive, greater exogenous capability raises early seller participation, reduces late entry, and nevertheless expands the period-2 seller base. Buyer

participation rises in both periods, more strongly in period 1, so subsequent buyer growth narrows. These joint movements distinguish front-loading from simple intertemporal substitution: fewer late entrants need not imply a smaller continuation market. Seller patience generates a separate, nonmonotone prediction. It reduces early and period-2 seller participation when the early cohort is below half the seller mass required to close late entry, but increases them once the cohort exceeds that threshold; late entry moves in the opposite direction. Tests of this reversal should interact a waiting-value shifter with a measure of the installed seller base rather than estimate a single average effect of patience.

The pricing predictions provide further testable implications. Within the positive-late-entry region, greater capability lowers both the period-1 seller fee and the late-entrant fee, raises the incumbent continuation fee, and widens the incumbent–entrant fee gap by less than one for one. The first two responses reflect recruitment of higher-friction sellers, whereas the incumbent fee capitalizes acquired capability. The platform can charge a positive period-1 seller fee only in this region and only below the subsidy threshold. If the fee begins positive, it becomes a subsidy before late entry disappears; after late entry closes, the early-seller fee is always negative. At the closure boundary, the waiting-rent concession is zero, so the remaining subsidy isolates the seller-to-buyer revenue motive. These predictions should not be tested using posted commission rates alone when rebates or grants vary across cohorts. In addition, the incumbent-entrant fee gap is well defined only while late entry is positive; once the entrant fee is unused, its equilibrium level is set-valued.

Commitment. Mapping an empirical treatment to the full-commitment benchmark requires that the platform be allowed to choose and credibly commit, before seller entry, to the complete history-contingent continuation schedule; imposing an arbitrary future fee does not implement the benchmark. Zero late entry alone does not identify commitment because limited commitment also produces three zero-late-entry regions. Relative to limited commitment, full commitment raises early recruitment when $\mathcal{V} < \mathcal{V}_x$, yields the same early recruitment at $\mathcal{V} = \mathcal{V}_x$, lowers it when $\mathcal{V}_x < \mathcal{V} < \mathcal{V}_2$, and yields the same allocation again when $\mathcal{V} \geq \mathcal{V}_2$. Under common discounting and unrestricted transfers, whenever commitment changes the allocation, it raises platform profit but reduces period-2 participation on both sides and lowers social welfare; both regimes choose the same support intensity. Fee levels are a weaker comparison because full commitment permits payments to be shifted across dates without changing their present value. Empirical work should emphasize

participation, present-value cohort payments, profit, and social welfare rather than individual dated fees.

Managerial implications. Platforms should manage acquisition transfers, capability expenditures, and cohort activity as distinct objects. A useful operating dashboard would separately report the net transfer to each entry cohort, real support per early seller, early active sellers, incumbent retention, late activation, and the period-2 seller and buyer bases. This decomposition prevents two common errors: treating a transfer as a resource cost and interpreting a decline in late entry as evidence of weaker market development. It also helps identify which type of region the market falls under. Within limited commitment, a locally flat nonsaturated seller base may reflect the waiting-option plateau, whereas an expanding zero-late-entry seller base indicates the interior zero-late-entry region; at saturation, additional capability cannot expand the seller side. Under full commitment, by contrast, zero late entry and equal seller masses across periods obtain throughout the nonsaturated solution.

Commitment should likewise be evaluated against the platform’s objective. In the benchmark, allowing the platform to choose and commit to its optimal continuation schedule raises profit whenever commitment changes the allocation, but the gain is accompanied by lower period-2 participation and social welfare. Thus, managers and policymakers should evaluate commitment devices jointly with cohort participation and cross-side market development, not infer their value from platform profit or the absence of late entry alone.

6 Conclusion

This paper studies how a platform combines seller acquisition pricing with costly capability support when sellers can delay entry and the platform cannot commit to continuation fees. Future entrant terms determine the value of waiting that must be compensated in period 1, while early participation creates a productive cohort that the platform can monetize in period 2. Under limited commitment, this interaction generates four regimes as the net return to early capability rises: positive late entry, a zero-late-entry plateau, renewed early recruitment without late entry, and seller saturation. The plateau arises because early recruitment lowers the waiting concession paid to the entire early cohort; when late entry closes, this cohort-wide marginal saving disappears. The platform nevertheless continues to subsidize early sellers at closure, isolating the seller-to-buyer revenue motive.

Capability and seller patience operate through distinct intertemporal margins. With positive late entry, greater capability front-loads seller participation and expands the period-2 market on both sides, even as late entry declines. Greater seller patience can either delay or accelerate recruitment, depending on the installed seller base. Capability also polarizes cohort fees by lowering early-seller and late-entrant fees while raising the incumbent continuation fee. These results show why entry flows, active seller stocks, real support, and net cohort payments must be observed jointly: no single measure identifies the direction or scale of market development.

Full commitment eliminates late entry but does not uniformly expand early recruitment. Under common discounting, whenever commitment changes the allocation, it raises platform profit while reducing period-2 participation and social welfare. Both regimes choose the same support intensity, which is conditionally socially optimal; the welfare ranking therefore reflects the timing and scale of participation rather than inefficient support. Commitment resolves the platform’s time inconsistency, but its private value can be accompanied by weaker market development and lower social welfare. The Online Appendix examines the robustness of the mechanism to alternative type distributions, pricing restrictions, imperfect enforcement, experience-dependent buyer returns, and AI-enabled support.

References

- Acquisti, A. and Varian, H. R. (2005). Conditioning prices on purchase history. *Marketing Science*, 24(3):367–381.
- AliResearch (2015). 2015 China Taobao Village research report. Technical report, AliResearch. In Chinese; [source link](#).
- Anderson, Jr., E. G., Parker, G. G., and Tan, B. (2014). Platform performance investment in the presence of network externalities. *Information Systems Research*, 25(1):152–172.
- Anderson, Jr., E. G., Parker, G. G., and Tan, B. (2023). Strategic investments for platform launch and ecosystem growth: A dynamic analysis. *Journal of Management Information Systems*, 40(3):807–839.
- Armstrong, M. (2006). Competition in two-sided markets. *RAND Journal of Economics*, 37(3):668–691.
- Arrow, K. J. (1962). The economic implications of learning by doing. *Review of Economic Studies*, 29(3):155–173.
- Bakos, Y. and Halaburda, H. (2022). Overcoming the coordination problem in new marketplaces via cryptographic tokens. *Information Systems Research*, 33(4):1368–1385.
- Cabral, L. (2019). Towards a theory of platform dynamics. *Journal of Economics & Management Strategy*, 28(1):60–72.
- Caillaud, B. and Jullien, B. (2003). Chicken & egg: Competition among intermediation service providers. *RAND Journal of Economics*, 34(2):309–328.
- Chu, J. and Manchanda, P. (2016). Quantifying cross and direct network effects in online consumer-to-consumer platforms. *Marketing Science*, 35(6):870–893.
- Couture, V., Faber, B., Gu, Y., and Liu, L. (2021). Connecting the countryside via E-Commerce: Evidence from China. *American Economic Review: Insights*, 3(1):35–50.
- de Cornière, A., Mantovani, A., and Shekhar, S. (2025). Third-degree price discrimination in two-sided markets. *Management Science*, 71(4):3340–3356.
- Dou, Y. and Wu, D. J. (2021). Platform competition under network effects: Piggybacking and optimal subsidization. *Information Systems Research*, 32(3):820–835.
- Fan, J., Tang, L., Zhu, W., and Zou, B. (2018). The Alibaba effect: Spatial consumption inequality and the welfare gains from E-Commerce. *Journal of International Economics*, 114:203–220.
- Hagiu, A. (2006). Pricing and commitment by two-sided platforms. *RAND Journal of Economics*, 37(3):720–737.

- Huang, Y. (2020). Alibaba: Online sales in 832 poverty-stricken counties reached RMB 97.4 billion in 2019. China.org.cn; reproduced by China News Service. In Chinese; [source link](#).
- Jeon, D.-S., Kim, B.-C., and Menicucci, D. (2022). Second-degree price discrimination by a two-sided monopoly platform. *American Economic Journal: Microeconomics*, 14(2):322–369.
- Li, J., Pisano, G., Xu, Y., and Zhu, F. (2023). Marketplace scalability and strategic use of platform investment. *Management Science*, 69(7):3958–3975.
- Lin, S. (2020). Two-sided price discrimination by media platforms. *Marketing Science*, 39(2):317–338.
- Liu, Q. and Serfes, K. (2013). Price discrimination in two-sided markets. *Journal of Economics & Management Strategy*, 22(4):768–786.
- Luo, X. and Niu, C. (2019). E-Commerce participation and household income growth in Taobao Villages. Policy Research Working Paper 8811, World Bank.
- Ministry of Commerce of the People’s Republic of China (2017). Rural online retail sales reached RMB 894.54 billion in 2016. Department of Market System Development. In Chinese; [source link](#).
- Ministry of Commerce of the People’s Republic of China (2020). E-Commerce in China 2019. Technical report, Ministry of Commerce of the People’s Republic of China. [Source link](#).
- Ministry of Commerce of the People’s Republic of China (2022a). China had 7,023 Taobao Villages in 2021. Trade in Services Network. In Chinese; [source link](#).
- Ministry of Commerce of the People’s Republic of China (2022b). E-Commerce in China 2021. Technical report, Ministry of Commerce of the People’s Republic of China. [Source link](#).
- Parker, G. G. and Van Alstyne, M. W. (2005). Two-sided network effects: A theory of information product design. *Management Science*, 51(10):1494–1504.
- Qi, J., Zheng, X., and Guo, H. (2019). The formation of Taobao villages in China. *China Economic Review*, 53:106–127.
- Rochet, J.-C. and Tirole, J. (2003). Platform competition in two-sided markets. *Journal of the European Economic Association*, 1(4):990–1029.
- Rochet, J.-C. and Tirole, J. (2006). Two-sided markets: A progress report. *RAND Journal of Economics*, 37(3):645–667.
- Spence, A. M. (1981). The learning curve and competition. *Bell Journal of Economics*, 12(1):49–70.
- Tan, B., Anderson, Jr., E. G., and Parker, G. G. (2020). Platform pricing and investment to drive third-party value creation in two-sided networks. *Information Systems Research*, 31(1):217–239.

- Villas-Boas, J. M. (2004). Price cycles in markets with customer recognition. *RAND Journal of Economics*, 35(3):486–501.
- Weyl, E. G. (2010). A price theory of multi-sided platforms. *American Economic Review*, 100(4):1642–1672.
- World Bank and Alibaba Group (2019). E-Commerce development: Experience from China. Technical report, World Bank, Washington, DC. [Source link](#).
- Zhou, Y. (2017). Bayesian estimation of a dynamic model of two-sided markets: Application to the U.S. video game industry. *Management Science*, 63(11):3874–3894.

Online Appendix to “Join or Wait? Seller Capability Development and Dynamic Platform Pricing”

Kevin Mi, Danxia Xie, Buyuan Yang, and Hanzhe Zhang

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1 Institutional Figure and Data

1.1 Data Sources and Construction of Figure 1

Definitions. AliResearch classifies a rural administrative village as a Taobao Village when annual e-commerce sales reach RMB 10 million and either at least 100 active online shops operate in the village or active shops account for at least 10 percent of local households. The count is therefore not the number of Rural Taobao service stations. Beginning with the 2020 report, AliResearch introduced an agricultural-sales adjustment: agricultural-product sales count twice toward the RMB 10 million threshold. The 2020 count is consequently not definition-homogeneous with earlier counts.

From 2016 onward, the Ministry of Commerce defines national rural online retail sales as sales to individuals by e-commerce businesses, including individual operators, that operate at the county level or below and outside urban districts. This is a seller-location measure. By contrast, the 2014 value of RMB 180 billion was originally reported as an AliResearch estimate of the rural online-shopping market, and the 2015 observation was reported as RMB 353 billion in rural-area online-shopping transactions. Later Ministry releases reproduce 180 as the 2014 historical headline, but that retrospective presentation does not eliminate differences in the descriptions and statistical bases of the underlying observations. We therefore preserve the published 180–2,050 sequence shown in the source manuscripts but do not present it as a constant-definition growth series.

Table OA.1: Value-by-Value Sources: National Series

Year	Rural sales: source value	Sales source	Taobao Villages	Village-count source
2014	RMB 180 bn (estimate)	AliResearch, 《农村电子商务消费报告 (2014)》 [<i>Rural E-Commerce Consumption Report 2014</i>]; later Ministry retrospective	212	AliResearch historical series in 《中国淘宝村研究报告 (2016)》 [<i>China Taobao Village Research Report (2016)</i>]
2015	RMB 353 bn	Ministry of Commerce transcript of the State Council policy briefing	780	Contemporaneous Xinhua report on AliResearch’s 《2015 年中国淘宝村研究报告》 [<i>2015 China Taobao Village Research Report</i>]

Continued on next page

Table OA.1 continued

Year	Rural sales: source value	Sales source	Taobao Villages	Village-count source
2016	RMB 894.54 bn	Ministry of Commerce release and definition	1,311	AliResearch, 《中国淘宝村研究报告 (2016)》 [<i>China Taobao Village Research Report (2016)</i>]
2017	RMB 1,244.88 bn	Ministry of Commerce release	2,118	World Bank–Alibaba report, reproducing the AliResearch series
2018	RMB 1,368 bn	Ministry release (RMB 1.37 tn) and 《中国电子商务报告 2019》 [<i>E-Commerce in China 2019</i>]	3,202	World Bank–Alibaba report, reproducing the AliResearch series
2019	RMB 1,708 bn	《中国电子商务报告 2019》 [<i>E-Commerce in China 2019</i>] (release: RMB 1.7 tn)	4,310	AliResearch count from 《中国淘宝村研究报告 (2009–2019)》 [<i>China Taobao Village Research Report 2009–2019</i>], reproduced by the Ministry
2020	RMB 1,790 bn	Ministry of Commerce planning review	5,425	AliResearch, 《1% 的改变——2020 中国淘宝村研究报告》 [<i>One Percent Makes a Difference: 2020 China Taobao Village Research Report</i>]
2021	RMB 2,050 bn	《中国农村电子商务发展报告 (2021–2022)》 [<i>China Rural E-Commerce Development Report 2021–2022</i>]	7,023	Ministry release citing the AliResearch list

Reconciliation of the national village counts. The contemporaneous 2015 AliResearch report records 780 Taobao Villages in 2015, which is the value used here. The two prior manuscript reproductions supplied for this audit display 778 and 779, respectively; we treat those values as transcription or source-vintage differences rather than overriding the contemporaneous report. The same AliResearch report records the consolidated 2014 count of 212, although some early announcements reported 211. The 2020 report both supplies the 5,425 count and documents the new agricultural-sales weighting rule described above.

Designated counties and platform coverage. The designated-county panels use the historical list of 832 counties in the national poverty-alleviation campaign: 592 Key Counties for National Poverty Alleviation and Development plus 240 additional counties in the 14 contiguous destitute areas. The same World Bank–Alibaba report states that its transaction-data extract for 2013–2016 covers 828 of the 832 counties because four observations were unavailable; the published 2016 aggregate is retained without imputation. All 832 counties had exited the official designation by November 2020, so the panels end in 2019 and the label denotes historical, not current, status.

The county sales observations are seller sales on Alibaba’s retail platforms, principally Taobao and Tmall. They do not cover all platforms and are not directly comparable with the national series. The original disclosures for 2014, 2017, and 2018 say “more than” RMB 12, 37, and 63 billion. Figure 1 plots the disclosed thresholds, not estimated exact values. In 2015, RMB 21.556 billion is rounded to RMB 22 billion; in 2019, RMB 97.4 billion is rounded to RMB 97 billion. The last observation is reported in Alibaba’s fiscal-year-2020 public-welfare report and is also described in contemporaneous coverage as the 2019 value; the figure follows the latter year label rather than asserting a more precise transaction window.

Table OA.2: Value-by-Value Sources: Designated-County Series

Year	Alibaba sales: source value	Sales source	Taobao Villages	Village-count source
2014	More than RMB 12 bn	Alibaba, 《2014 年县域暨农村电商数据》 [<i>2014 County and Rural E-Commerce Data</i>]	4	Retrospective series in 《1%的改变——2020 中国淘宝村研究报告》 [<i>One Percent Makes a Difference: 2020 China Taobao Village Research Report</i>]
2015	RMB 21.556 bn	《2015 年中国县域电子商务报告》 [<i>2015 China County E-Commerce Report</i>], exact value reproduced by the World Bank	10	Retrospective series in 《1%的改变——2020 中国淘宝村研究报告》 [<i>One Percent Makes a Difference: 2020 China Taobao Village Research Report</i>]
2016	RMB 29 bn	World Bank–Alibaba, <i>E-Commerce Development: Experience from China</i>	18	Retrospective series in 《1%的改变——2020 中国淘宝村研究报告》 [<i>One Percent Makes a Difference: 2020 China Taobao Village Research Report</i>]
2017	More than RMB 37 bn	《新乡村新消费新商业——农村商业研究报告》 [<i>New Countryside, New Consumption, New Business: Rural Commerce Research Report</i>]	33	World Bank–Alibaba report
2018	More than RMB 63 bn	《阿里巴巴脱贫工作报告 2018》 [<i>Alibaba Poverty Alleviation Work Report 2018</i>], reproduced by the Cyberspace Administration of China	45	World Bank–Alibaba report
2019	RMB 97.4 bn	Alibaba, 《阿里巴巴 2020 财年公益“财报”》 [<i>Alibaba Fiscal Year 2020 Public-Welfare “Financial Report”</i>]; contemporaneous 2019-year coverage	63	《中国淘宝村研究报告 (2009–2019)》 [<i>China Taobao Village Research Report 2009–2019</i>] summary

2 Limited-Commitment Results

2.1 No Planned Incumbent Exit

Lemma 2.1 (No planned incumbent exit). *At an optimal period-1 policy, every early seller remains active in period 2: $n_{2I}^{s*} = n_1^{s*}$. Thus, the platform never deliberately recruits a positive mass of sellers that it subsequently excludes.*

For this proof only, let

$$s \equiv \alpha + \beta, \quad Q \equiv 4 - s^2, \quad H \equiv 2 - s^2. \quad (\text{OA.2.1})$$

These symbols do not enter the main statements.

Proof. Continuation problem.

Fix an inherited early-seller mass $n_1^s \in (0, 1]$ and support $x \geq 0$. Let b , r , and e denote period-2 buyers, retained incumbents, and late entrants. The period-2 objective is

$$\Pi_2(b, r, e) = b - b^2 + sb(r + e) + (c + x)r - r^2 - n_1^s e - e^2. \quad (\text{OA.2.2})$$

Its Hessian is

$$\begin{pmatrix} -2 & s & s \\ s & -2 & 0 \\ s & 0 & -2 \end{pmatrix}, \quad (\text{OA.2.3})$$

which is negative definite for $0 < s < 1$. The KKT allocation is therefore unique and globally optimal.

Candidate partial-retention continuations.

Solving the KKT system yields the following two candidate activity sets. Partial incumbent retention and positive late entry occur if and only if

$$c + x < \frac{s}{H} \quad (\text{OA.2.4})$$

and

$$\frac{2s + Q(c + x)}{8 - 3s^2} < n_1^s < \frac{2s + s^2(c + x)}{Q}. \quad (\text{OA.2.5})$$

In this case,

$$n_2^b = \frac{2 + s(c + x - n_1^s)}{2H}, \quad (\text{OA.2.6})$$

$$n_{2I}^s = \frac{2s + Q(c + x) - s^2 n_1^s}{4H}, \quad (\text{OA.2.7})$$

$$n_{2E}^s = \frac{2s + s^2(c + x) - Qn_1^s}{4H}. \quad (\text{OA.2.8})$$

Partial incumbent retention and zero late entry occur if and only if

$$n_1^s > \frac{s + 2(c + x)}{Q} \quad \text{and} \quad n_1^s \geq \frac{s[2 + s(c + x)]}{Q}. \quad (\text{OA.2.9})$$

In this case,

$$n_2^b = \frac{2 + s(c + x)}{Q}, \quad n_{2I}^s = \frac{s + 2(c + x)}{Q}, \quad n_{2E}^s = 0. \quad (\text{OA.2.10})$$

With positive late entry, implementing fees satisfy

$$p_2^b = 1 + \beta(n_{2I}^s + n_{2E}^s) - n_2^b, \quad (\text{OA.2.11})$$

$$p_{2I}^s = \alpha n_2^b + c + x - n_{2I}^s, \quad (\text{OA.2.12})$$

$$p_{2E}^s = \alpha n_2^b - n_1^s - n_{2E}^s. \quad (\text{OA.2.13})$$

With zero late entry, the first two equations continue to apply, and any $p_{2E}^s \geq \alpha n_2^b - n_1^s$ deters entry. Imposing $0 < r < n_1^s$ together with $e > 0$ or $e = 0$, respectively, yields the conditions displayed above.

Excluding planned exit.

In the partial-retention/zero-late-entry activity set, the derivative of reduced period-1 profit with respect to n_1^s , holding x fixed, is

$$\frac{\partial \Pi}{\partial n_1^s} = \frac{s}{2} - \frac{Qn_1^s}{2} - \frac{\kappa x^2}{2} < 0, \quad (\text{OA.2.14})$$

where the inequality follows from the first condition in (OA.2.9).

In the partial-retention/positive-late-entry activity set, the envelope theorem gives

$$\frac{\partial \Pi}{\partial n_1^s} = \frac{s}{2} - \frac{Qn_1^s}{2} - (\delta_p + \delta_s)n_{2E}^s + \frac{\delta_s Qn_1^s}{4H} - \frac{\kappa x^2}{2}. \quad (\text{OA.2.15})$$

This derivative is affine in n_1^s . To display the endpoint signs, write $g = c + x$, $R = 8 - 3s^2$, and let

$$n_L = \frac{2s + Qg}{R}, \quad n_U = \frac{2s + s^2g}{Q} \quad (\text{OA.2.16})$$

denote the two boundaries in (OA.2.5). Substitution gives

$$\begin{aligned} \left. \frac{\partial \Pi}{\partial n_1^s} \right|_{n_1^s = n_L} &= - \frac{2s^3(1 - s^2) + g(32 - 40s^2 + 15s^4 - 2s^6)}{4HR} \\ &\quad - \frac{(1 - \delta_s)[2s^3 + g(32 - 24s^2 + 5s^4)]}{4HR} - \frac{\delta_p(s - Hg)}{R} - \frac{\kappa x^2}{2} < 0, \end{aligned} \quad (\text{OA.2.17})$$

and

$$\left. \frac{\partial \Pi}{\partial n_1^s} \right|_{n_1^s = n_U} = -\frac{s\{2(H-1) + sg(2H-1) + (1-\delta_s)(2+sg)\}}{4H} - \frac{\kappa x^2}{2} < 0. \quad (\text{OA.2.18})$$

Every term is weakly negative, and at least one is strictly negative, because $0 < s < 1$, $\delta_s < 1$, and $g < s/H$ in this activity set. Because the derivative is affine in n_1^s and negative at both endpoints, it is negative throughout the interval. A small reduction in early recruitment is therefore profitable in either partial-retention activity set, proving the lemma. $\square$

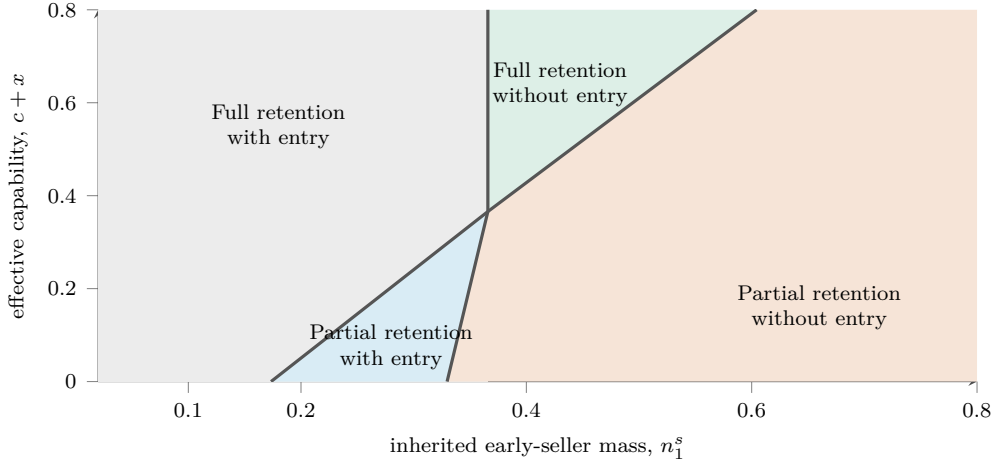


Figure OA.1: Period-2 Continuations after an Inherited Seller Base

Notes. The figure fixes $\alpha + \beta = 0.60$. The horizontal axis is the inherited early-seller mass n_1^s , and the vertical axis is effective capability $c + x$. The shaded regions are determined by the exact KKT conditions derived in this subsection. Strict interiors are locally robust continuations; adjacent activity sets share their boundaries. The figure is a continuation map, not an on-path global-equilibrium phase diagram.

2.2 Proof of Theorem 1

Proof. Period-2 continuation.

Fix n_1^s and x . Substituting the period-2 inverse demands (7)–(9) into period-2 revenue gives

$$\begin{aligned} \Pi_2 = & n_2^b - (n_2^b)^2 + (\alpha + \beta)n_2^b(n_{2I}^s + n_{2E}^s) \\ & + (c + x)n_{2I}^s - (n_{2I}^s)^2 - n_1^s n_{2E}^s - (n_{2E}^s)^2. \end{aligned} \quad (\text{OA.2.19})$$

The Hessian with respect to $(n_2^b, n_{2I}^s, n_{2E}^s)$ is

$$\begin{pmatrix} -2 & \alpha + \beta & \alpha + \beta \\ \alpha + \beta & -2 & 0 \\ \alpha + \beta & 0 & -2 \end{pmatrix}. \quad (\text{OA.2.20})$$

The leading principal minors alternate in sign when $(\alpha + \beta)^2 < 2$, so the continuation objective is strictly concave. Assumption 1 implies this condition.

By Lemma 2.1, the on-path continuation has full retention, $n_{2I}^s = n_1^s$. The first-order conditions for buyers and late entrants are

$$1 - 2n_2^b + (\alpha + \beta)(n_1^s + n_{2E}^s) = 0, \quad (\text{OA.2.21})$$

$$(\alpha + \beta)n_2^b - n_1^s - 2n_{2E}^s = 0. \quad (\text{OA.2.22})$$

Solving gives (10) and (12). The full-retention KKT condition is

$$(\alpha + \beta)n_2^{b*} + c + x - 2n_1^s \geq 0. \quad (\text{OA.2.23})$$

We verify this KKT condition at the period-1 optimum below.

The positive-late-entry branch.

The marginal early seller's continuation payoffs imply (13) and (14). Optimizing period-1 buyer revenue gives

$$n_1^{b*} = \frac{1 + (\alpha + \beta)n_1^s}{2}, \quad p_1^{b*} = \frac{1 + (\beta - \alpha)n_1^s}{2}. \quad (\text{OA.2.24})$$

After these substitutions, platform profit in the positive-late-entry region is

$$\begin{aligned} \Pi(n_1^s, x) = & \frac{[1 + (\alpha + \beta)n_1^s]^2}{4} - (n_1^s)^2 \\ & - \delta_s n_1^s \frac{(\alpha + \beta) - [2 - (\alpha + \beta)^2]n_1^s}{4 - (\alpha + \beta)^2} - \frac{\kappa x^2}{2} n_1^s \\ & + \delta_p \left[(c + x)n_1^s - (n_1^s)^2 + \frac{1 + (\alpha + \beta)n_1^s + (n_1^s)^2}{4 - (\alpha + \beta)^2} \right]. \end{aligned} \quad (\text{OA.2.25})$$

For $n_1^s > 0$, the support first-order condition is $n_1^s(\delta_p - \kappa x) = 0$, so $x^* = \delta_p/\kappa$. The first-order condition for n_1^s is linear; after substituting x^* , its solution is the Region (i) early-seller mass in Theorem 1. The denominator of that solution is positive because

$$[4 - (\alpha + \beta)^2]^2 - 4\delta_s[2 - (\alpha + \beta)^2] + 4\delta_p[3 - (\alpha + \beta)^2] > 0 \quad (\text{OA.2.26})$$

under $0 < \alpha + \beta < 1$, $\delta_s < 1$, and $\delta_p > 0$. Up to a positive factor, the left-hand side of (OA.2.26) is also the negative of the second derivative of (OA.2.25) with respect to n_1^s , so the first-order condition is sufficient. The Region (i) parameter condition is equivalent to $n_1^{s*} < (\alpha + \beta)/[2 - (\alpha + \beta)^2]$, and hence to $n_{2E}^{s*} > 0$. Substituting the allocation into the inverse demands yields the fees in Proposition 1.

Global regime characterization.

When late entry is positive, optimizing x and the buyer and entrant quantities reduces platform profit to

$$\begin{aligned} \Pi^+(n_1^s) = & \frac{[1 + (\alpha + \beta)n_1^s]^2}{4} - (n_1^s)^2 - \delta_s n_1^s \frac{(\alpha + \beta) - [2 - (\alpha + \beta)^2]n_1^s}{4 - (\alpha + \beta)^2} \\ & + \left(\delta_p c + \frac{\delta_p^2}{2\kappa} \right) n_1^s + \delta_p \left[-(n_1^s)^2 + \frac{1 + (\alpha + \beta)n_1^s + (n_1^s)^2}{4 - (\alpha + \beta)^2} \right]. \end{aligned} \quad (\text{OA.2.27})$$

This branch is feasible when

$$n_1^s < \frac{\alpha + \beta}{2 - (\alpha + \beta)^2}. \quad (\text{OA.2.28})$$

When late entry is zero, the optimized buyer mass is $[1 + (\alpha + \beta)n_1^s]/2$ and the reduced profit is

$$\Pi^0(n_1^s) = \frac{1 + \delta_p}{4} [1 + (\alpha + \beta)n_1^s]^2 - (1 + \delta_p)(n_1^s)^2 + \left(\delta_p c + \frac{\delta_p^2}{2\kappa} \right) n_1^s. \quad (\text{OA.2.29})$$

This branch is feasible when the reverse weak inequality holds in (OA.2.28).

The two profit functions coincide at

$$n_1^s = \frac{\alpha + \beta}{2 - (\alpha + \beta)^2}. \quad (\text{OA.2.30})$$

The derivative approaching this point from the positive-late-entry side is

$$\delta_p c + \frac{\delta_p^2}{2\kappa} - \frac{(1 + \delta_p)(\alpha + \beta)}{2 - (\alpha + \beta)^2} + \frac{\delta_s(\alpha + \beta)}{4 - (\alpha + \beta)^2}, \quad (\text{OA.2.31})$$

whereas the derivative from the zero-late-entry side is

$$\delta_p c + \frac{\delta_p^2}{2\kappa} - \frac{(1 + \delta_p)(\alpha + \beta)}{2 - (\alpha + \beta)^2}. \quad (\text{OA.2.32})$$

The derivative therefore falls by $\delta_s(\alpha + \beta)/[4 - (\alpha + \beta)^2]$ when the late-entry margin becomes inactive. Each branch is strictly concave in n_1^s , and the derivative jumps downward at their common boundary. The piecewise objective is thus globally strictly concave in the early-seller mass.

If the left derivative (OA.2.31) is negative, the unique maximizer lies on the positive-late-entry branch. If the left derivative is weakly positive and the right derivative (OA.2.32) is weakly negative, the unique maximizer is the kink (OA.2.30). If the right derivative is positive, the unique interior maximizer lies on the zero-late-entry branch and equals the Region (iii) seller mass in Theorem 1. That mass reaches one at the Region (iv) threshold. These cases establish the four-region characterization and continuity at the boundaries.

Full retention.

It remains to verify the full-retention KKT condition at the optimum. In the positive-late-entry region, $c + x^* \geq (\delta_p c + \delta_p^2/(2\kappa))/\delta_p$. After substituting the positive-late-entry solution, the resulting

lower bound on the left-hand side of (OA.2.23) is affine in $\delta_p c + \delta_p^2/(2\kappa)$. At zero net capability return, it equals

$$\frac{(\alpha + \beta) \{ [4 - (\alpha + \beta)^2](\alpha + \beta)^2 + 2\delta_p[4 - (\alpha + \beta)^2] + 2\delta_s(\alpha + \beta)^2 \}}{[4 - (\alpha + \beta)^2] \{ [4 - (\alpha + \beta)^2]^2 - 4\delta_s[2 - (\alpha + \beta)^2] + 4\delta_p[3 - (\alpha + \beta)^2] \}} > 0. \quad (\text{OA.2.33})$$

At the upper boundary of the positive-late-entry region, it is bounded below by

$$\frac{\alpha + \beta}{\delta_p} \left[\frac{1}{2 - (\alpha + \beta)^2} - \frac{\delta_s}{4 - (\alpha + \beta)^2} \right] > 0. \quad (\text{OA.2.34})$$

Because this lower bound is affine in the net capability return and is positive at both endpoints of Region (i), it is positive throughout that region. On the plateau, $c + x^* > n_1^{s*}$, which is sufficient for full retention. At an interior zero-late-entry optimum, the KKT condition reduces to $c + x^* \geq [\delta_p c + \delta_p^2/(2\kappa)]/(1 + \delta_p)$ and therefore holds; the same argument applies at seller saturation.

Implementation uniqueness.

Fix (p_1^b, p_1^s, x) . Whenever buyer participation is interior, $n_1^b = 1 + \beta n_1^s - p_1^b$. For the full-retention/positive-late-entry, zero-late-entry, and partial-retention/positive-late-entry activity sets, respectively, the derivative of the marginal seller's implementing payoff with respect to n_1^s is

$$\alpha\beta - 1 + \delta_s \frac{2 - (\alpha + \beta)^2}{4 - (\alpha + \beta)^2}, \quad \alpha\beta - 1, \quad \alpha\beta - 1 + \delta_s \frac{4 - (\alpha + \beta)^2}{4[2 - (\alpha + \beta)^2]}. \quad (\text{OA.2.35})$$

Each expression is strictly negative: $\alpha\beta < 1/4$, the fraction in the full-retention/positive-late-entry slope is below 1/2, the corresponding fraction in the partial-retention/positive-late-entry slope is below 3/4, and $\delta_s < 1$. The marginal seller's implementing payoff is continuous at activity-set boundaries and strictly decreasing over the full piecewise continuation map. A fixed fee vector therefore implements at most one early-seller mass, after which buyer participation is unique. Together with Lemma 2.1, these arguments complete the proof. $\square$

2.3 Proof of Proposition 1

Proof. Theorem 1 determines n_1^{s*} , n_{2E}^{s*} , $n_{2I}^{s*} = n_1^{s*}$, and $x^* = \delta_p/\kappa$ in every region. Given an early-seller mass n_1^{s*} , substituting the period-1 inverse demands into platform revenue and maximizing over buyers gives

$$n_1^{b*} = \frac{1 + (\alpha + \beta)n_1^{s*}}{2}, \quad p_1^{b*} = \frac{1 + (\beta - \alpha)n_1^{s*}}{2}. \quad (\text{OA.2.36})$$

The marginal early seller's lifetime indifference condition then gives the period-1 seller-fee formula in Proposition 1.

In Region (i), the positive-late-entry branch derived in the proof of Theorem 1 gives (10)–(12). Substituting these quantities into the inverse demands (7)–(9) gives the three continuation prices reported in the proposition.

In Regions (ii)–(iv), Theorem 1 gives $n_{2E}^{s*} = 0$ and a common seller mass $n_2^{s*} = n_1^{s*}$. Optimizing buyer participation at that seller mass gives

$$n_2^{b*} = n_1^{b*} = \frac{1 + (\alpha + \beta)n_1^{s*}}{2}. \quad (\text{OA.2.37})$$

The buyer inverse demand implies $p_2^{b*} = p_1^{b*}$, and the incumbent inverse demand implies $p_{2I}^{s*} = p_1^{s*} + c + x^*$. Substituting the plateau, interior, and saturation values of n_1^{s*} yields the corresponding formulas in the proposition.

Any $p_{2E}^{s*} \geq \alpha n_2^{b*} - n_1^{s*}$ deters late entry. This lower bound is the stated entry-detering threshold in Regions (ii)–(iv). The allocation and every fee paid in equilibrium are therefore unique. The entrant fee is posted on the equilibrium path but is not paid; any fee at or above the threshold implements the same allocation, so this unused fee is set-valued. $\square$

2.4 Proof of Corollary 1

Proof. Fix κ and substitute $\delta_p = \kappa x^*$ into the net capability return:

$$\delta_p c + \frac{\delta_p^2}{2\kappa} = \kappa x^* c + \frac{\kappa(x^*)^2}{2} = \kappa x^* \left(c + \frac{x^*}{2} \right). \quad (\text{OA.2.38})$$

Equating (OA.2.38) to the three regime thresholds in Theorem 1 and solving for c gives

$$c_1(x^*) = \frac{\alpha + \beta}{2 - (\alpha + \beta)^2} + \frac{1}{\kappa x^*} \left[\frac{\alpha + \beta}{2 - (\alpha + \beta)^2} - \frac{\delta_s(\alpha + \beta)}{4 - (\alpha + \beta)^2} \right] - \frac{x^*}{2}, \quad (\text{OA.2.39})$$

$$c_2(x^*) = \frac{\alpha + \beta}{2 - (\alpha + \beta)^2} + \frac{\alpha + \beta}{\kappa x^* [2 - (\alpha + \beta)^2]} - \frac{x^*}{2}, \quad (\text{OA.2.40})$$

$$c_3(x^*) = \frac{4 - (\alpha + \beta)^2 - (\alpha + \beta)}{2} + \frac{4 - (\alpha + \beta)^2 - (\alpha + \beta)}{2\kappa x^*} - \frac{x^*}{2}. \quad (\text{OA.2.41})$$

The regime inequalities follow because the left side of (OA.2.38) is strictly increasing in c .

Let b_j denote the coefficient multiplying $1/(\kappa x^*)$ in $c_j(x^*)$. Assumption 1 implies $b_j > 0$ for every $j \in \{1, 2, 3\}$. The only non-immediate case is

$$b_1 = \frac{\alpha + \beta}{2 - (\alpha + \beta)^2} - \frac{\delta_s(\alpha + \beta)}{4 - (\alpha + \beta)^2} > 0.$$

Therefore

$$\frac{dc_j}{dx^*} = -\frac{b_j}{\kappa(x^*)^2} - \frac{1}{2} < 0, \quad \frac{d^2 c_j}{d(x^*)^2} = \frac{2b_j}{\kappa(x^*)^3} > 0, \quad j \in \{1, 2, 3\}. \quad (\text{OA.2.42})$$

Subtracting the first boundary from the second gives

$$c_2(x^*) - c_1(x^*) = \frac{\delta_s(\alpha + \beta)}{\kappa x^* [4 - (\alpha + \beta)^2]} = \frac{\delta_s(\alpha + \beta)}{\delta_p [4 - (\alpha + \beta)^2]}. \quad (\text{OA.2.43})$$

Writing $s = \alpha + \beta$, $Q = 4 - s^2$, and $H = 2 - s^2$, direct subtraction also gives

$$c_3(x^*) - c_2(x^*) = \left(1 + \frac{1}{\kappa x^*}\right) \frac{(1-s)(2-s)(s+2)^2}{2H} > 0. \quad (\text{OA.2.44})$$

Together, the two differences verify $c_1(x^*) < c_2(x^*) < c_3(x^*)$. The network-direction statements used in the main-text discussion also follow directly. With $a = 1/(\kappa x^*)$,

$$\begin{aligned} \frac{\partial c_2}{\partial s} &= (1+a) \frac{2+s^2}{(2-s^2)^2} > 0, & \frac{\partial c_3}{\partial s} &= -(1+a) \frac{1+2s}{2} < 0, \\ \frac{\partial c_1}{\partial s} &= \frac{2+s^2}{(2-s^2)^2} + a \left[\frac{2+s^2}{(2-s^2)^2} - \delta_s \frac{4+s^2}{(4-s^2)^2} \right] > 0. \end{aligned}$$

For the last inequality, the expression in brackets is minimized at $\delta_s = 1$; after this substitution, the numerator of the difference is $2(8 + 6s^2 - 3s^4) > 0$. Finally, if instead δ_p is fixed while κ varies, then $x^* = \delta_p/\kappa$ and the net return is $\delta_p(c + x^*/2)$. Equating it to any fixed regime threshold gives a line of slope -2 in (c, x^*) space. $\square$

2.5 Supplementary Regime Figures

The main text reports the capability–network map, which most directly addresses the paper’s market-development question. The figures below show how platform patience, parameterized by the induced support level $x^* = \delta_p/\kappa$, interacts with capability and cross-side effects. With κ fixed, a horizontal movement changes both the platform’s weight on continuation profit and its optimal support; it is therefore not a pure support shock.

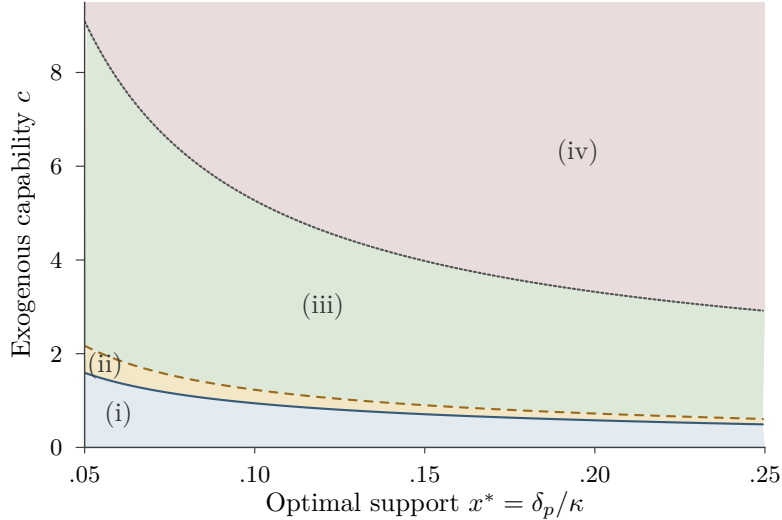


Figure OA.2: Capability, Platform Patience, and Equilibrium Regimes

Notes. The figure fixes $\alpha + \beta = 0.60$, $\delta_s = 0.70$, and $\kappa = 4$. The horizontal axis reports $x^* = \delta_p/\kappa$, so $x^* \leq 0.25$; the plotted range begins at 0.05 for readability. Regions (i)–(iv) denote positive late entry, the zero-late-entry plateau, the interior zero-late-entry regime, and seller saturation. The solid, dashed, and dotted curves are the three analytical boundaries in Theorem 1.

For fixed x^* , greater capability moves the equilibrium through Regions (i)–(iv). Moving right raises both the platform’s weight on continuation profit and its optimal support because κ is fixed. All three capability thresholds fall, while the plateau narrows according to (OA.2.43).

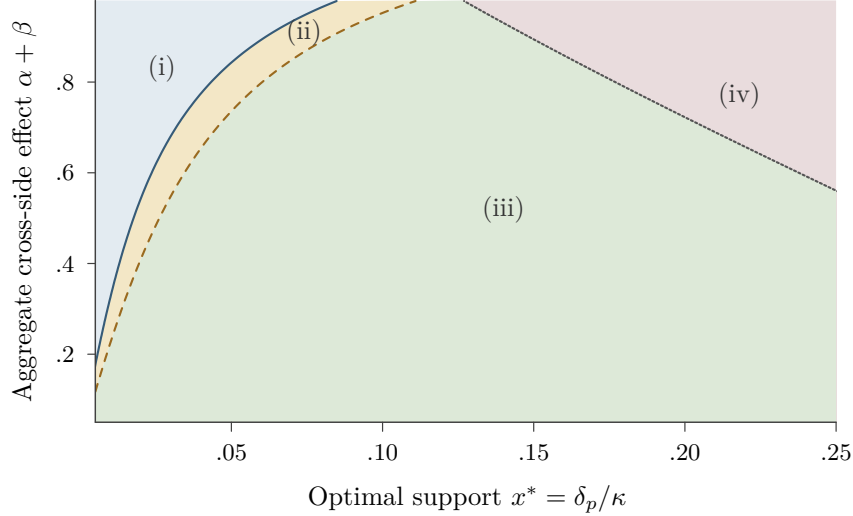


Figure OA.3: Cross-Side Effects, Platform Patience, and Equilibrium Regimes

Notes. The figure fixes $c = 3$, $\delta_s = 0.70$, and $\kappa = 4$. The horizontal axis reports $x^* = \delta_p/\kappa$, and the vertical axis reports $\alpha + \beta \in (0, 1)$. The left endpoint represents the limit $x^* \downarrow 0$; tick marks begin at 0.05 for readability. Regions (i)–(iv) correspond to the four cases in Theorem 1. The boundary curves are evaluated directly from the theorem’s analytical conditions.

A larger $\alpha + \beta$ raises both the value of an installed seller base and the threshold $\bar{n} = (\alpha + \beta)/[2 - (\alpha + \beta)^2]$ at which the residual entrant pool is exhausted. The first force strengthens early acquisition; the second requires a larger initial cohort before late entry vanishes. The nonparallel boundaries reflect these opposing margins.

2.6 Proof of Proposition 2

Proof. For this proof, let

$$s = \alpha + \beta, \quad Q = 4 - s^2, \quad H = 2 - s^2,$$

and let

$$Z = Q^2 - 4\delta_s H + 4\delta_p(3 - s^2) > 0.$$

In Region (i), Theorem 1 and Proposition 1 imply

$$n_1^{s*} = \frac{sQ + 2s(\delta_p - \delta_s) + Q(2\delta_p c + \delta_p^2/\kappa)}{Z}, \quad p_1^{s*} = \frac{\alpha}{2} - \frac{\delta_s s}{Q} + \left(\frac{\alpha s}{2} - 1 + \frac{\delta_s H}{Q}\right) n_1^{s*}. \quad (\text{OA.2.45})$$

Hence the fee is affine in c , with

$$\frac{\partial p_1^{s*}}{\partial c} = \frac{\delta_p \{Q(\alpha s - 2) + 2\delta_s H\}}{Z} < 0. \quad (\text{OA.2.46})$$

The inequality follows because $Q(\alpha s - 2) + 2\delta_s H < -Q + 2H = -s^2 < 0$. At the boundary where late entry closes, $n_1^{s*} = s/H$ and $n_{2E}^{s*} = 0$, so

$$p_1^{s*} = -\frac{\beta}{H} < 0. \quad (\text{OA.2.47})$$

Strict monotonicity and affineness therefore give a unique zero c^S , possibly below the feasible capability range, and $p_1^{s*} < 0$ exactly when $c > c^S$ within Region (i).

On the Region (ii) plateau, the fee equals the boundary value in (OA.2.47). In Region (iii), $n_1^{s*} > s/H$ and

$$p_1^{s*} = \frac{\alpha}{2} + \left(\frac{\alpha s}{2} - 1\right) n_1^{s*}.$$

Because $\alpha s/2 - 1 < 0$, the fee is strictly below its value at $n_1^{s*} = s/H$, namely $-\beta/H$. In Region (iv),

$$p_1^{s*} = \frac{\alpha(1+s)}{2} - 1 < 0,$$

because $\alpha < 1$ and $s < 1$. This proves the result in all four regions. $\square$

2.7 Primitive Subsidy Threshold

The proof of Proposition 2 establishes that the Region (i) fee is affine and strictly decreasing in c . For a compact primitive expression, let

$$s = \alpha + \beta, \quad Q = 4 - s^2, \quad H = 2 - s^2, \quad Z = Q^2 - 4\delta_s H + 4\delta_p(3 - s^2), \quad A = \frac{\alpha s}{2} - 1 + \frac{\delta_s H}{Q}. \quad (\text{OA.2.48})$$

Solving the affine fee in (OA.2.45) gives

$$c^S = -\frac{Z\left(\frac{\alpha}{2} - \frac{\delta_s s}{Q}\right) + A\left[sQ + 2s(\delta_p - \delta_s) + Q\frac{\delta_p^2}{\kappa}\right]}{2\delta_p Q A}. \quad (\text{OA.2.49})$$

Here $A < 0$ because $\alpha s/2 < 1/2$ and $H/Q < 1/2$. Hence the denominator is nonzero, and (OA.2.49) is the unique affine zero characterized in Proposition 2.

2.8 Proof of Proposition 3

Proof. Seller patience. Differentiating the Region (i) early-seller formula in Theorem 1 gives

$$\frac{\partial n_1^{s*}}{\partial \delta_s} = \frac{2\{2[2 - (\alpha + \beta)^2]n_1^{s*} - (\alpha + \beta)\}}{[4 - (\alpha + \beta)^2]^2 - 4\delta_s[2 - (\alpha + \beta)^2] + 4\delta_p[3 - (\alpha + \beta)^2]}. \quad (\text{OA.2.50})$$

Because the denominator and $2 - (\alpha + \beta)^2$ are positive, the sign of (OA.2.50) is the sign of $n_1^{s*} - \bar{n}/2$. Moreover, differentiating $n_{2E}^{s*} = \{(\alpha + \beta) - [2 - (\alpha + \beta)^2]n_1^{s*}\}/[4 - (\alpha + \beta)^2]$ and using $n_2^{s*} = n_1^{s*} + n_{2E}^{s*}$ gives

$$\frac{\partial n_{2E}^{s*}}{\partial \delta_s} = -\frac{2 - (\alpha + \beta)^2}{4 - (\alpha + \beta)^2} \frac{\partial n_1^{s*}}{\partial \delta_s}, \quad \frac{\partial n_2^{s*}}{\partial \delta_s} = \frac{2}{4 - (\alpha + \beta)^2} \frac{\partial n_1^{s*}}{\partial \delta_s}. \quad (\text{OA.2.51})$$

These identities establish all seller-patience signs in Proposition 3. On the Region (ii) plateau, Theorem 1 pins n_1^{s*} at $\bar{n}$. The Region (iii) allocation does not depend on δ_s , and seller saturation makes every participation quantity locally constant. *Capability.* Differentiating the Region (i) formula for n_1^{s*} gives $\partial n_1^{s*}/\partial c > 0$. The identities $n_{2E}^{s*} = \{(\alpha + \beta) - [2 - (\alpha + \beta)^2]n_1^{s*}\}/[4 - (\alpha + \beta)^2]$ and $n_2^{s*} = n_1^{s*} + n_{2E}^{s*}$ then establish the signs of the late-entry and period-2 seller responses. Let $s = \alpha + \beta$ and $Q = 4 - s^2$. The buyer responses satisfy

$$\frac{\partial n_1^{b*}}{\partial c} = \frac{s}{2} \frac{\partial n_1^{s*}}{\partial c} > \frac{s}{Q} \frac{\partial n_1^{s*}}{\partial c} = \frac{\partial n_2^{b*}}{\partial c} > 0,$$

where $Q > 2$. Finally, $n_2^{b*} - n_1^{b*} = sn_{2E}^{s*}/2$ implies that the buyer-participation gap narrows. On the plateau, every capability derivative is zero. In Region (iii), the two seller masses have the same positive derivative with respect to c , and the two buyer masses likewise have the same positive derivative. At seller saturation, all participation derivatives are zero. $\square$

2.9 Proof of Corollary 2

Proof. Using the Region (i) fees in Proposition 1,

$$\frac{\partial p_1^{s*}}{\partial c} = \left[\frac{\alpha(\alpha + \beta)}{2} - 1 + \frac{\delta_s[2 - (\alpha + \beta)^2]}{4 - (\alpha + \beta)^2} \right] \frac{\partial n_1^{s*}}{\partial c} < 0, \quad (\text{OA.2.52})$$

$$\frac{\partial p_{2E}^{s*}}{\partial c} = \frac{\alpha(\alpha + \beta) - 2}{4 - (\alpha + \beta)^2} \frac{\partial n_1^{s*}}{\partial c} < 0, \quad (\text{OA.2.53})$$

$$\frac{\partial p_{2I}^{s*}}{\partial c} = 1 - \left[1 - \frac{\alpha(\alpha + \beta)}{4 - (\alpha + \beta)^2} \right] \frac{\partial n_1^{s*}}{\partial c} > 0. \quad (\text{OA.2.54})$$

The inequalities use $0 < \alpha + \beta < 1$, $\delta_s < 1$, and $0 < \partial n_1^{s*}/\partial c < 1$. To verify the last bound, let $s = \alpha + \beta$, $Q = 4 - s^2$, $H = 2 - s^2$, and $Z = Q^2 - 4\delta_s H + 4\delta_p(3 - s^2)$. Then

$$\frac{\partial n_1^{s*}}{\partial c} = \frac{2\delta_p Q}{Z} < 1,$$

because $Z - 2\delta_p Q = Q^2 - 4\delta_s H + 2\delta_p H > Q^2 - 4H = H^2 + 4 > 0$. Subtracting the entrant fee from the incumbent fee in Proposition 1 proves (16), which is strictly positive. Moreover,

$$\frac{\partial(p_{2I}^{s*} - p_{2E}^{s*})}{\partial c} = 1 - \frac{H}{Q} \frac{\partial n_1^{s*}}{\partial c} \in (0, 1),$$

because $0 < H/Q < 1$ and $0 < \partial n_1^{s*}/\partial c < 1$. This proves the remaining claims. $\square$

3 Full-Commitment Results

3.1 Proof of Proposition 4

Proof. Let $s = \alpha + \beta$, $Q = 4 - s^2$, and $\mathcal{V} = \delta c + \delta^2/(2\kappa)$. For $n > 0$, optimize support and hold n and the period-2 seller mass $m = r + e$ fixed. The derivative with respect to retention is

$$\delta c + 2\delta(n - r) + 2\delta(m - r) + \frac{\delta^2 r}{\kappa n} > 0$$

whenever $r < n$ and $e > 0$, so partial retention with positive late entry cannot be optimal. Now suppose $e = 0$ and $r < n$. If $r < s/Q$, increasing retention raises $\phi(r)$ because $\phi'(r) > 0$ and also weakly raises the capability contribution. If $r \geq s/Q$, reducing early recruitment from n to r raises current revenue because ϕ is decreasing above s/Q , lowers support cost, and leaves continuation revenue unchanged. Thus, every optimum has $r = n$.

With full retention, the support condition gives $x = \delta/\kappa$, and the problem reduces to

$$\max_{0 \leq n \leq m \leq 1} \{\phi(n) + \delta\phi(m) + \mathcal{V}n\}.$$

The unconstrained targets are $(s + 2\mathcal{V})/Q$ for n and s/Q for m . Because $\mathcal{V} > 0$, the former is strictly larger, so strict concavity implies that $n \leq m$ binds. Solving the resulting one-dimensional problem yields

$$n = m = \min \left\{ 1, \frac{(1 + \delta)s + 2\mathcal{V}}{(1 + \delta)Q} \right\}.$$

The derivative at $n = m = 0$ is positive, so the solution has $n > 0$ and the interior support condition applies.

Finally, the allocation is implemented by

$$\begin{aligned} b_1 &= \frac{1 + sn}{2}, & b_2 &= \frac{1 + s(r + e)}{2}, \\ p_1^s &= \alpha b_1 - n - \delta e, & p_{2I}^s &= \alpha b_2 + c + x - r, & p_{2E}^s &= \alpha b_2 - n - e. \end{aligned} \quad (\text{OA.3.1})$$

At the optimum, $r = n = m$ and $e = 0$. These fees support the stated buyer and seller masses. The unused entrant fee may take any value weakly above the entry-detering threshold, and common discounting otherwise leaves the intertemporal division of incumbent payments nonunique. $\square$

3.2 Proof of Proposition 5

Proof. Let $s = \alpha + \beta$, $Q = 4 - s^2$, $H = 2 - s^2$, and $\mathcal{V} = \delta c + \delta^2/(2\kappa)$. Under limited commitment and $\mathcal{V} < \mathcal{V}_1$, early recruitment is

$$n_1^{s,L} = \frac{Q(s + 2\mathcal{V})}{Q^2 + 4\delta}, \quad (\text{OA.3.2})$$

whereas full commitment gives

$$n_1^{s,C} = \frac{(1 + \delta)s + 2\mathcal{V}}{(1 + \delta)Q}. \quad (\text{OA.3.3})$$

Their difference is

$$n_1^{s,L} - n_1^{s,C} = \frac{2\delta(6-s^2)H}{(1+\delta)Q(Q^2+4\delta)}(\mathcal{V} - \mathcal{V}_\times). \quad (\text{OA.3.4})$$

The ordering follows from positivity of the thresholds and

$$\begin{aligned} \mathcal{V}_2 - \mathcal{V}_1 &= \frac{\delta s}{Q} > 0, \\ \mathcal{V}_1 - \mathcal{V}_\times &= \frac{s(Q^2+4\delta)}{QH(6-s^2)} > 0. \end{aligned} \quad (\text{OA.3.5})$$

Thus $0 < \mathcal{V}_\times < \mathcal{V}_1 < \mathcal{V}_2$. Limited commitment selects $n_1^{s,L} = s/H$ on $[\mathcal{V}_1, \mathcal{V}_2]$ and the same zero-late-entry allocation as full commitment at and above $\mathcal{V}_2$. For $\mathcal{V} < \mathcal{V}_1$, direct substitution gives

$$n_2^{s,L} - n_2^{s,C} = \frac{2\{(1+\delta)Qs - H(Q-2\delta)\mathcal{V}\}}{(1+\delta)Q(Q^2+4\delta)} > 0.$$

The numerator vanishes at $(1+\delta)Qs/[H(Q-2\delta)]$, which exceeds $\mathcal{V}_2$ and therefore exceeds every $\mathcal{V} < \mathcal{V}_1$. For $\mathcal{V} \in [\mathcal{V}_1, \mathcal{V}_2)$, limited commitment gives $n_2^{s,L} = s/H$, whereas the full-commitment mass remains strictly below s/H . Buyer participation is strictly increasing in the contemporaneous seller mass, so the same strict ranking holds for period-2 buyers. These comparisons establish the period-2 allocation rankings.

Full commitment expands the set of feasible price plans. The limited-commitment plan is feasible under full commitment and reproduces Π^L . Whenever $\mathcal{V} < \mathcal{V}_2$, however, it does not induce the unique full-commitment allocation; strict concavity therefore implies $\Pi^C > \Pi^L$. For $\mathcal{V} \geq \mathcal{V}_2$, the two regimes select the same allocation from the same reduced objective, $(1+\delta)\phi(n) + \mathcal{V}n$, including the common seller-saturation corner. Hence $\Pi^C = \Pi^L$ on this range. $\square$

3.3 Proof of Proposition 6

Proof. Let $s = \alpha + \beta$, $Q = 4 - s^2$, $H = 2 - s^2$, and $\mathcal{V} = \delta c + \delta^2/(2\kappa)$. One-period social welfare at seller mass z is the sum of buyer surplus, seller surplus, and platform profit, net of seller operating costs; access fees cancel as transfers. Evaluating this sum at the equilibrium buyer mass $[1 + sz]/2$ gives

$$\omega(z) = \frac{3 + 6sz + (3s^2 - 4)z^2}{8}. \quad (\text{OA.3.6})$$

For early seller mass n and period-2 seller mass m , discounted welfare is

$$W(n, m) = \omega(n) + \delta\omega(m) + \mathcal{V}n. \quad (\text{OA.3.7})$$

In the positive-late-entry region, the limited-commitment period-2 seller mass is $n_2^{s,L} = n_1^{s,L} + n_{2E}^{s,L}$, with $n_{2E}^{s,L}$ given by Theorem 1. Substituting this mass and (OA.3.2) into (OA.3.7) gives

$$W^L - W^C = \frac{\delta\{(1+\delta)s^2[12 - 5s^2 - 8s\mathcal{V}] + [144 - 108s^2 + 20s^4 - s^6]\mathcal{V}^2\}}{2(1+\delta)Q^2(Q^2+4\delta)} > 0. \quad (\text{OA.3.8})$$

To verify the sign, the numerator in braces is a quadratic in $\mathcal{V}$ with positive leading and constant coefficients. Its discriminant is

$$4(1 + \delta)s^2[-1728 + 2016s^2 - (764 - 16\delta)s^4 + 112s^6 - 5s^8] < 0. \quad (\text{OA.3.9})$$

For $0 < s < 1$ and $0 < \delta < 1$, the bracket is increasing in both s^2 and δ . Its upper bound at $s^2 = \delta = 1$ is -353 , so the discriminant is negative and the quadratic is strictly positive. On the plateau, the difference is

$$\frac{[(1 + \delta)s - H\mathcal{V}] \{ \mathcal{V}(24 - 14s^2 + s^4) + (1 + \delta)s(12 - 5s^2) \}}{2(1 + \delta)Q^2H^2}. \quad (\text{OA.3.10})$$

This expression is strictly positive for $\mathcal{V} \in [\mathcal{V}_1, \mathcal{V}_2)$ and equals zero at $\mathcal{V} = \mathcal{V}_2$. At and above $\mathcal{V}_2$, the allocations coincide, so welfare is equal. $\square$

4 Robustness and Extensions

4.1 General Type Distributions: Cutoff Strategies and Continuation Uniqueness

The main text obtains closed-form results under uniform type distributions. The cutoff formulation and a sufficient uniqueness condition extend beyond the uniform baseline to settings with a seller-friction distribution F_s with positive density f_s and a buyer-value distribution F_b with positive density f_b on compact supports. Let q_1 denote the period-1 seller cutoff. Conditional on early participation, the incumbent seller distribution is

$$F_I(\theta \mid q_1) = \begin{cases} F_s(\theta)/F_s(q_1), & \theta \leq q_1, \\ 1, & \theta > q_1, \end{cases} \quad (\text{OA.4.1})$$

whereas the distribution among potential late entrants is

$$F_E(\theta \mid q_1) = \begin{cases} 0, & \theta \leq q_1, \\ [F_s(\theta) - F_s(q_1)]/[1 - F_s(q_1)], & \theta > q_1. \end{cases} \quad (\text{OA.4.2})$$

These posteriors separate the capability effect of early participation from the selection effect generated by the cutoff.

Given participation quantities, the period-2 marginal seller cutoffs are $\alpha n_2^b + c + x - p_{2I}^s$ for incumbents and $\alpha n_2^b - p_{2E}^s$ for entrants. Buyer participation satisfies

$$n_t^b = 1 - F_b(p_t^b - \beta n_t^s). \quad (\text{OA.4.3})$$

The period-1 gain from joining rather than waiting is the general-distribution counterpart of (5). Differentiating this gain with respect to seller friction θ gives a derivative no greater than $-(1 - \delta_s) <$

0, because each continuation value is the maximum of zero and an affine function with slope -1 . Early participation therefore continues to satisfy single crossing.

Lemma 4.1 (Sufficient fulfilled-expectations uniqueness). *Let $\bar{f}_s = \sup_{\theta} f_s(\theta)$ and $\bar{f}_b = \sup_v f_b(v)$. For any fixed history and price vector, the period-2 fulfilled-expectations participation vector is unique if*

$$2\alpha\beta\bar{f}_s\bar{f}_b < 1.$$

Proof of Lemma 4.1. For fixed fees and a fixed period-1 history, the map from buyer participation to the two active seller cutoffs has slope α . Applying the density bound to both cohorts and then composing with buyer demand yields a buyer-to-seller-to-buyer map with Lipschitz modulus at most $2\alpha\beta\bar{f}_s\bar{f}_b$. The displayed condition makes this modulus strictly smaller than one, so the Banach fixed-point theorem gives a unique fulfilled-expectations participation vector. $\square$

This condition guarantees a unique participation fixed point, but does not by itself guarantee a unique dynamic platform optimum. The uniform baseline and Assumption 1 provide the stronger structure used for the global closed-form characterization.

4.2 Uniform Continuation Pricing

To separate commitment from cohort-based pricing, consider a supplementary benchmark in which the platform cannot condition the period-2 seller fee on participation history. Thus, $p_{2I}^s = p_{2E}^s \equiv p_2^s$, while the platform retains limited commitment and all other features of the baseline model. Superscript U denotes this pricing restriction.

Proposition 4.1 (Uniform continuation pricing). *Under Assumption 1, uniform period-2 seller pricing generates full retention and no late entry:*

$$n_{2I}^{s,U} = n_1^{s,U}, \quad n_{2E}^{s,U} = 0.$$

Support and early seller participation are

$$x^U = \frac{\delta_p}{\kappa}, \quad n_1^{s,U} = \min \left\{ 1, \frac{(1 + \delta_p)(\alpha + \beta) + 2\delta_p c + \delta_p^2/\kappa}{(1 + \delta_p)[4 - (\alpha + \beta)^2]} \right\}.$$

Buyer participation is constant across periods:

$$n_1^{b,U} = n_2^{b,U} = \frac{1 + (\alpha + \beta)n_1^{s,U}}{2}.$$

The allocation and all on-path prices are independent of seller patience.

Uniform pricing pools two endogenous cohorts. At a candidate allocation with n retained early sellers and $e > 0$ late entrants, the entrant cutoff determines the common fee. Relative to history-contingent pricing, capability and entrant selection are then left as incumbent rent: each installed

seller pays $c + x + e$ less, reducing continuation revenue by $n(c + x + e)$. The platform avoids this cohort-wide revenue loss by recruiting in period 1 every seller it plans to serve.

Uniform pricing and full commitment therefore generate the same entry-timing pattern under common discounting, but through different mechanisms. Uniform pricing links late-entrant acquisition to incumbent revenue; full commitment instead makes future entrant terms credible before sellers choose when to enter.

Proof. Fix n_1^s and x . If the common period-2 seller fee induces positive late entry, all incumbents are retained because they have both lower costs and the additional capability $c + x$. Let total period-2 seller participation be $n_1^s + n_{2E}^s$. The platform's period-2 profit becomes

$$n_2^b - (n_2^b)^2 + (\alpha + \beta)n_2^b(n_1^s + n_{2E}^s) - (n_1^s + n_{2E}^s)^2. \quad (\text{OA.4.4})$$

The expansion solution is

$$n_2^{b,U} = \frac{2}{4 - (\alpha + \beta)^2}, \quad (\text{OA.4.5})$$

$$n_1^s + n_{2E}^{s,U} = \frac{\alpha + \beta}{4 - (\alpha + \beta)^2}, \quad (\text{OA.4.6})$$

$$p_2^{s,U} = \frac{\alpha - \beta}{4 - (\alpha + \beta)^2}. \quad (\text{OA.4.7})$$

Its continuation profit is $1/[4 - (\alpha + \beta)^2]$. Because the marginal late entrant determines the common fee, acquired capability is left as incumbent rent.

Alternatively, the platform can set a higher common fee that serves incumbents while deterring late entrants:

$$p_2^s = \alpha \frac{1 + (\alpha + \beta)n_1^s}{2} + c + x - n_1^s \quad (\text{OA.4.8})$$

and obtains continuation profit

$$\frac{[1 + (\alpha + \beta)n_1^s]^2}{4} + (c + x)n_1^s - (n_1^s)^2. \quad (\text{OA.4.9})$$

The expansion branch dominates the incumbent-only branch if and only if

$$\frac{4 - (\alpha + \beta)^2}{4} \left[\frac{\alpha + \beta}{4 - (\alpha + \beta)^2} - n_1^s \right]^2 > (c + x)n_1^s. \quad (\text{OA.4.10})$$

Suppose, toward a contradiction, that an on-path history satisfies (OA.4.10). Under expansion, the marginal early seller values capability by $c + x$, so the implementing period-1 fee is

$$p_1^s = \alpha n_1^b - n_1^s + \delta_s(c + x). \quad (\text{OA.4.11})$$

Consider instead recruiting $(\alpha + \beta)/[4 - (\alpha + \beta)^2]$ sellers in period 1, choosing $x = 0$, and monetizing them in period 2. Relative to the candidate expansion path, the profit gain is

$$\begin{aligned} & \frac{1}{4 - (\alpha + \beta)^2} - \left\{ \frac{[1 + (\alpha + \beta)n_1^s]^2}{4} - (n_1^s)^2 \right\} \\ & - \delta_s(c + x)n_1^s + \frac{\kappa x^2}{2}n_1^s + \delta_p c \frac{\alpha + \beta}{4 - (\alpha + \beta)^2}. \end{aligned} \quad (\text{OA.4.12})$$

Condition (OA.4.10) implies that this gain is strictly greater than

$$(1 - \delta_s)(c + x)n_1^s + \frac{\kappa x^2}{2}n_1^s + \delta_p c \frac{\alpha + \beta}{4 - (\alpha + \beta)^2} \geq 0. \quad (\text{OA.4.13})$$

Every positive-late-entry path is therefore strictly dominated.

A partial-retention path is also dominated. The unconstrained retained mass in the incumbent-monetization branch is $[\alpha + \beta + 2(c + x)]/[4 - (\alpha + \beta)^2]$, which is weakly greater than the static seller optimum $(\alpha + \beta)/[4 - (\alpha + \beta)^2]$. Partial retention requires early recruitment above this level, where first-period platform revenue is strictly decreasing. Reducing early recruitment to the retained mass raises current profit, lowers support cost, and leaves continuation profit unchanged.

The resulting on-path problem therefore has full retention and zero late entry. After optimizing buyer participation, it is

$$\max_{0 \leq n_1^s \leq 1, x \geq 0} \left\{ \frac{1 + \delta_p}{4} [1 + (\alpha + \beta)n_1^s]^2 - (1 + \delta_p)(n_1^s)^2 + \delta_p(c + x)n_1^s - \frac{\kappa x^2}{2}n_1^s \right\}. \quad (\text{OA.4.14})$$

Strict concavity gives $x^U = \delta_p/\kappa$ and the seller mass stated in Proposition 4.1, with the unit upper bound producing the minimum operator. Buyer participation follows from inverse demand. Because the waiting option is inactive, δ_s does not enter (OA.4.14). $\square$

4.3 Imperfect Enforceability of History-Contingent Seller Pricing

Suppose an incumbent can reset its observable seller identity before period 2. Resetting preserves a fraction $\rho \in [0, 1]$ of the capability created by early participation and entails a private cost $d \geq 0$. If the seller keeps its incumbent identity, its period-2 payoff is

$$\alpha n_2^b - \theta + c + x - p_{2I}^s. \quad (\text{OA.4.15})$$

If it resets and claims the entrant price, its payoff is

$$\alpha n_2^b - \theta + \rho(c + x) - p_{2E}^s - d. \quad (\text{OA.4.16})$$

Proposition 4.2 (Imperfect enforceability of history-contingent seller pricing). *History-contingent seller prices must satisfy*

$$p_{2I}^s - p_{2E}^s \leq (1 - \rho)(c + x) + d.$$

Conditional on full incumbent retention and positive late entry, the unconstrained history-contingent price vector is enforceable if and only if

$$n_{2E}^s \leq d - \rho(c + x).$$

When the constraint binds, the marginal incumbent earns reset rent

$$[n_{2E}^s + \rho(c + x) - d]_+,$$

and, for positive late entry, the first-period seller fee becomes

$$p_1^s = \alpha n_1^b - n_1^s + \delta_s \left([n_{2E}^s + \rho(c + x) - d]_+ - n_{2E}^s \right).$$

Thus identity resetting raises the first-period seller fee relative to the fully enforceable baseline by $\delta_s [n_{2E}^s + \rho(c + x) - d]_+$.

If $\delta_p \geq \delta_s$, no interior on-path equilibrium can feature both a strictly binding enforcement constraint and positive late entry. Under this discounting condition, partial enforceability therefore yields three candidate on-path outcomes: the unconstrained history-contingent equilibrium, an enforcement-boundary allocation satisfying

$$n_{2E}^s = d - \rho(c + x),$$

or zero late entry. In the polar case $\rho = 1$ and $d = 0$, late entry is zero and the equilibrium allocation coincides with that under history-neutral seller pricing. As $d \rightarrow \infty$, the baseline history-contingent equilibrium is recovered.

The result characterizes when history-contingent seller prices are enforceable. Identity resetting narrows the feasible incumbent–entrant fee gap and weakens the need for an early monetary subsidy: future reset rent is capitalized into the period-1 fee.

Proof of Proposition 4.2. Write $s = \alpha + \beta$, $Q = 4 - s^2$, $H = 2 - s^2$, $n = n_1^s$, $e = n_{2E}^s$, and $g = c + x$. Comparing (OA.4.15) and (OA.4.16) gives the enforceability constraint displayed in Proposition 4.2. Under the unconstrained baseline, the fee gap is $g + e$ by (16); hence the constraint is slack if and only if $g + e \leq (1 - \rho)g + d$, which is the proposition’s displayed slack condition.

The entrant price that implements $e > 0$ is

$$p_{2E}^s = \alpha n_2^b - n - e. \tag{OA.4.17}$$

If the constraint binds, the highest enforceable incumbent price is

$$p_{2I}^s = \alpha n_2^b + g - n - [e + \rho g - d]_+. \tag{OA.4.18}$$

The marginal incumbent therefore earns the reset rent displayed in Proposition 4.2. Leaving this rent through a lower incumbent fee is weakly more profitable than inducing actual identity resetting, so an indifferent seller retains its incumbent identity. Comparing the marginal seller's incumbent and waiting values in period 1 yields the displayed period-1 fee.

It remains to characterize a strictly binding positive-late-entry interior. After optimizing the buyer fee, buyer participation is

$$n_2^b = \frac{1 + s(n + e)}{2}. \quad (\text{OA.4.19})$$

When the constraint is slack, period-2 profit as a function of e is

$$\Pi_2^D(e) = \frac{[1 + s(n + e)]^2}{4} + gn - n^2 - ne - e^2, \quad (\text{OA.4.20})$$

whose interior maximizer is

$$e_D(n) = \frac{s - Hn}{Q}. \quad (\text{OA.4.21})$$

When the constraint strictly binds, period-2 profit is

$$\Pi_2^R(e) = \frac{[1 + s(n + e)]^2}{4} - (n + e)^2 + [(1 - \rho)g + d]n, \quad (\text{OA.4.22})$$

and its interior entrant quantity is

$$e_R(n) = \frac{s}{Q} - n. \quad (\text{OA.4.23})$$

Thus binding enforceability creates the same installed-base cost as a uniform seller price.

Let

$$w = \delta_p(1 - \rho) + \delta_s \rho. \quad (\text{OA.4.24})$$

On the strictly binding branch, the period-1 fee displayed in Proposition 4.2, support capitalization, and the quadratic cost give the interior support candidate $x^R = w/\kappa$. After optimizing the period-1 buyer fee, the early seller candidate is

$$n^R = \frac{s + 2Z_R}{Q}, \quad Z_R = wc + (\delta_p - \delta_s)d + \frac{w^2}{2\kappa}. \quad (\text{OA.4.25})$$

Substitution into (OA.4.23) gives

$$e^R = -\frac{2Z_R}{Q}. \quad (\text{OA.4.26})$$

If $\delta_p \geq \delta_s$, then $Z_R > 0$ because $c, d \geq 0$, $\kappa > 0$, and $w > 0$. Equation (OA.4.26) is then negative, contradicting strictly positive entry. The only remaining positive-late-entry candidates are a slack allocation or the boundary $e = d - \rho g$; otherwise late entry is zero.

If $\rho = 1$ and $d = 0$, the boundary requires $e = -g$, which is infeasible for $e > 0$, and the slack condition is also infeasible. The allocation therefore has zero late entry. The enforceability condition requires $p_{2I}^s \leq p_{2E}^s$, so the platform can implement the history-neutral allocation by choosing an unused entrant fee weakly above the incumbent fee in its on-path price schedule.

Finally, as $d \rightarrow \infty$ the constraint is slack at every finite baseline price vector, which recovers the unconstrained model. $\square$

4.4 Experience-Dependent Buyer-Base Absorption

In the additive baseline, $c + x$ shifts an experienced seller's gross surplus without altering period-2 quantity first-order conditions conditional on the installed base. We now allow experience to change how effectively a seller uses the platform's buyer base. Replace the incumbent's period-2 payoff by

$$u_{i2,I}^s = \lambda \alpha n_2^b - \theta_i + c + x - p_{2I}^s, \quad \lambda \geq 1. \quad (\text{OA.4.27})$$

Here λ is a structural learning parameter, not an algebraic shorthand. The term $(\lambda - 1)\alpha n_2^b$ captures experience with traffic tools, search placement, advertising, analytics, and customer conversion that makes the same buyer base more valuable to an incumbent. The case $\lambda = 1$ recovers the main model exactly.

We report results on an admissible interval of λ for which $2 - (\alpha + \beta)(\beta + \lambda\alpha) > 0$, full incumbent retention remains optimal, and the stated participation quantities are interior. For an inherited early seller mass N on the positive-late-entry continuation branch, direct optimization gives

$$\begin{aligned} n_2^b(N, \lambda) &= \frac{2 + [2(\beta + \lambda\alpha) - (\alpha + \beta)]N}{4 - (\alpha + \beta)^2}, \\ n_{2E}^s(N, \lambda) &= \frac{\alpha + \beta - \{2 - (\alpha + \beta)(\beta + \lambda\alpha)\}N}{4 - (\alpha + \beta)^2}. \end{aligned} \quad (\text{OA.4.28})$$

The late-entry mass reaches zero at

$$\bar{N}(\lambda) = \frac{\alpha + \beta}{2 - (\alpha + \beta)(\beta + \lambda\alpha)}, \quad \frac{d\bar{N}}{d\lambda} = \frac{\alpha(\alpha + \beta)^2}{[2 - (\alpha + \beta)(\beta + \lambda\alpha)]^2} > 0, \quad (\text{OA.4.29})$$

$$p_{2I}^s - p_{2E}^s = c + x + (\lambda - 1)\alpha n_2^b + n_{2E}^s. \quad (\text{OA.4.30})$$

Thus the incumbent-entrant fee gap contains direct capability, buyer-base absorption, and endogenous selection.

Proposition 4.3 (Buyer-base absorption and the timing of market building). *Suppose the equilibrium lies strictly inside the full-retention, positive-late-entry activity set and $\delta_p \geq \delta_s$. On every admissible λ interval,*

$$\frac{dn_1^{s*}}{d\lambda} > 0, \quad \frac{d(n_1^{s*} + n_{2E}^{s*})}{d\lambda} > 0, \quad \frac{dp_1^{s*}}{d\lambda} < 0, \quad x^* = \frac{\delta_p}{\kappa}.$$

The response of late entry is generally ambiguous. Specifically,

$$\frac{dn_{2E}^{s*}}{d\lambda} > 0 \iff \frac{dn_1^{s*}}{d\lambda} < \frac{\alpha(\alpha + \beta)n_1^{s*}}{2 - (\alpha + \beta)(\beta + \lambda\alpha)}.$$

The ambiguity reflects two equilibrium forces. The *future-market expansion effect* raises continuation buyer demand and hence the payoff from late entry. The *front-loading effect* makes an early seller more valuable to the platform, inducing a lower acquisition fee and a larger installed base before capability is realized; this larger base depletes the residual entrant pool. Stronger absorptive learning therefore expands the period-2 seller base but can either increase subsequent entry or shift it toward period 1. Additive capability differs because, conditional on N , it has no direct effect on continuation quantities.

Corollary 4.1 (Buyer-base absorption and the zero-late-entry plateau). *Let $\mathcal{V} = \delta_p c + \delta_p^2/(2\kappa)$. On an admissible interval in which the positive-late-entry and zero-late-entry branches meet at $\bar{N}(\lambda)$ and no other corner binds, let $\mathcal{V}_1(\lambda)$ and $\mathcal{V}_2(\lambda)$ denote the lower and upper boundaries of the zero-late-entry plateau. Then*

$$\mathcal{V}_2(\lambda) - \mathcal{V}_1(\lambda) = \frac{\delta_s(\alpha + \beta)}{4 - (\alpha + \beta)^2},$$

and, for $k \in \{1, 2\}$,

$$\text{sgn}\left(\frac{d\mathcal{V}_k}{d\lambda}\right) = \text{sgn}\left((\alpha + \beta)^2[4 - (\alpha + \beta)^2] - 4\delta_p[1 - (\alpha + \beta)^2]\right).$$

Hence both boundaries fall with λ when

$$\delta_p > \frac{(\alpha + \beta)^2[4 - (\alpha + \beta)^2]}{4[1 - (\alpha + \beta)^2]},$$

and rise when the inequality is reversed.

Absorptive learning shifts the plateau without changing its width in the uniform-linear benchmark. A larger λ raises the early-seller mass required to exhaust the entrant pool.

A sufficiently patient platform counteracts this force by front-loading recruitment to monetize the larger future incumbent value, so late entry reaches zero at a lower level of additive capability. When the platform is less patient, the expansion of future entry opportunities dominates, and eliminating late entry requires more additive capability.

These comparative statics are local because sufficiently large λ may activate buyer-saturation or partial-retention constraints, which require a separate global activity-set analysis.

Proof of Proposition 4.3 and Corollary 4.1. For this proof only, let

$$m = \alpha + \beta, \quad h = \beta + \lambda\alpha, \quad D = 4 - m^2, \quad N = n_1^s.$$

Conditional on full incumbent retention, the period-2 quantity objective on the positive-late-entry branch is

$$\begin{aligned} \Pi_2 = & n_2^b[1 + \beta(N + n_{2E}^s) - n_2^b] + N[(h - \beta)n_2^b + c + x - N] \\ & + n_{2E}^s[\alpha n_2^b - N - n_{2E}^s]. \end{aligned} \tag{OA.4.31}$$

Its Hessian with respect to (n_2^b, n_{2E}^s) is $\begin{pmatrix} -2 & m \\ m & -2 \end{pmatrix}$, which is negative definite because $0 < m < 1$. Solving the two first-order conditions gives

$$n_2^b = \frac{2 + (2h - m)N}{D}, \quad n_{2E}^s = \frac{m - (2 - mh)N}{D},$$

which proves (OA.4.28). Setting the second expression to zero gives (OA.4.29). Direct inverse demand also gives

$$p_{2I}^s - p_{2E}^s = c + x + (\lambda - 1)\alpha n_2^b + n_{2E}^s,$$

proving (OA.4.30).

The marginal early seller's incumbent continuation rent remains zero, whereas waiting yields n_{2E}^s . Consequently,

$$p_1^s = \alpha n_1^b - N - \delta_s n_{2E}^s, \quad n_1^b = \frac{1 + mN}{2}.$$

The support first-order condition is unchanged: an additional unit of x is capitalized into the incumbent fee for each early seller, so $\delta_p - \kappa x = 0$ and $x^* = \delta_p / \kappa$. Define the net additive capability value $\mathcal{V} = \delta_p c + \delta_p^2 / (2\kappa)$. Substituting the continuation solution into the period-1 objective gives the positive-late-entry candidate

$$N^*(h) = \frac{mD - 2\delta_s m + 2\delta_p(2h - m) + 2D\mathcal{V}}{D^2 + 4\delta_s(mh - 2) - 4\delta_p(h^2 - mh + m^2 - 3)}. \quad (\text{OA.4.32})$$

Let Z denote the denominator, which is positive on the stated interior activity set. Implicit differentiation of the first-order condition yields

$$\frac{dN^*}{dh} = \frac{4\{\delta_p + N^*[\delta_p(2h - m) - \delta_s m]\}}{Z} > 0. \quad (\text{OA.4.33})$$

The inequality follows because $h \geq m$ and $\delta_p \geq \delta_s$.

Total period-2 seller participation can be written as

$$N^* + n_{2E}^{s*} = \frac{m}{D} + \frac{2 - m^2 + mh}{D} N^*.$$

Both the direct derivative of total period-2 seller participation with respect to h and the coefficient multiplying dN^*/dh are positive. Since $dh/d\lambda = \alpha > 0$, this proves that early recruitment and total period-2 seller participation increase with λ . Moreover,

$$p_1^{s*} = \frac{\alpha}{2} - \frac{\delta_s m}{D} + \left[\frac{\alpha m}{2} - 1 + \frac{\delta_s(2 - mh)}{D} \right] N^*.$$

The bracketed coefficient is negative because $(2 - mh)/D \leq (2 - m^2)/D < 1/2$, and its derivative with respect to h is $-\delta_s m/D < 0$. Together with $dN^*/dh > 0$, this proves $dp_1^{s*}/d\lambda < 0$. Finally,

$$\frac{dn_{2E}^{s*}}{dh} = \frac{mN^* - (2 - mh)dN^*/dh}{D},$$

which is positive exactly under the condition in Proposition 4.3. This completes the proof of Proposition 4.3.

To prove Corollary 4.1, equate the positive-late-entry candidate in (OA.4.32) to $\bar{N} = m/(2-mh)$ and evaluate the right derivative of the zero-late-entry branch at the same allocation. The two capability thresholds are

$$\begin{aligned}\mathcal{V}_2(h) &= \frac{m(2 + mh - m^2) + 2\delta_p(2m - h)}{2(2 - mh)}, \\ \mathcal{V}_1(h) &= \mathcal{V}_2(h) - \frac{\delta_s m}{D}.\end{aligned}$$

Their difference proves the plateau-width formula in Corollary 4.1. Since $dh/d\lambda = \alpha$,

$$\frac{d\mathcal{V}_k}{d\lambda} = \frac{\alpha\{m^2 D - 4\delta_p(1 - m^2)\}}{2(2 - mh)^2}, \quad k \in \{1, 2\}, \quad (\text{OA.4.34})$$

which proves the boundary-shift result and the stated platform-patience threshold. $\square$

4.5 AI-Enabled Support and Matching

We model AI as an exogenous technology state $a \geq 0$ with two distinct channels. First, AI can increase the productivity of seller support:

$$g(x, a) = c + (1 + \eta a)x, \quad \eta \geq 0. \quad (\text{OA.4.35})$$

Second, AI can change how strongly buyers value seller participation:

$$\beta(a) = \beta_0 + \mu a. \quad (\text{OA.4.36})$$

A positive μ captures improved matching, recommendation, or demand discovery. A negative μ captures substitution: generative assistance, automated service, or standardized AI output can reduce the incremental value of seller variety. The admissible AI range satisfies

$$0 < \beta_0 + \mu a, \quad 0 < \alpha + \beta_0 + \mu a < 1. \quad (\text{OA.4.37})$$

Optimizing support gives

$$x^*(a) = \frac{\delta_p(1 + \eta a)}{\kappa}. \quad (\text{OA.4.38})$$

The corresponding discounted net capability value is

$$\mathcal{V}(a) = \delta_p c + \frac{\delta_p^2(1 + \eta a)^2}{2\kappa}, \quad \mathcal{V}'(a) = \frac{\delta_p^2 \eta(1 + \eta a)}{\kappa}. \quad (\text{OA.4.39})$$

The global four-region equilibrium continues to apply after replacing $\alpha + \beta$ by $\alpha + \beta(a)$ and the baseline net capability value by $\mathcal{V}(a)$.

Proposition 4.4 (AI-enabled support and market front-loading). *Suppose $\eta > 0$ and $\mu = 0$. Within the positive-late-entry region,*

$$\frac{dx^*}{da} > 0, \quad \frac{dn_1^{s*}}{da} > 0, \quad \frac{dn_{2E}^{s*}}{da} < 0,$$

and

$$\frac{d(n_1^{s*} + n_{2E}^{s*})}{da} > 0, \quad \frac{d(n_2^{b*} - n_1^{b*})}{da} < 0.$$

AI-enabled support also increases the gap between the net capability value and the positive-to-zero-late-entry threshold, moving the equilibrium toward that boundary.

Pure seller enablement therefore expands the period-2 seller base but compresses subsequent buyer growth. AI raises the return to acquiring sellers in period 1, so the platform front-loads recruitment. This reduces the pool of potential late entrants and slows buyer growth between periods even though the period-2 seller base is larger. An empirical analysis based only on post-adoption growth could therefore misclassify an expansionary AI technology as reducing market development.

Proposition 4.5 (AI matching, seller substitution, and market development). *Suppose $\eta > 0$, let $\beta(a) = \beta_0 + \mu a$, and suppose the baseline $a = 0$ lies strictly inside the positive-late-entry region. Evaluated at this baseline, there exist thresholds*

$$\mu_M < 0 < \mu_G < \mu_E$$

such that

$$\begin{aligned} \left. \frac{d(n_1^{s*} + n_{2E}^{s*})}{da} \right|_{a=0} > 0 &\iff \mu > \mu_M, \\ \left. \frac{d(n_2^{b*} - n_1^{b*})}{da} \right|_{a=0} > 0 &\iff \mu > \mu_G, \\ \left. \frac{dn_{2E}^{s*}}{da} \right|_{a=0} > 0 &\iff \mu > \mu_E. \end{aligned}$$

Consequently, a moderately substitutionary AI technology, $\mu \in (\mu_M, 0)$, can expand the period-2 seller base while compressing subsequent buyer growth. A sufficiently strong matching technology expands both the period-2 seller base and buyer growth, although late seller entry continues to decline for $\mu \in (\mu_G, \mu_E)$. Only sufficiently strong seller substitution, $\mu < \mu_M$, contracts the period-2 seller base.

There also exists $\mu_C > 0$ such that a marginal increase in the AI technology state at $a = 0$ advances the transition to zero late entry if and only if $\mu < \mu_C$. Strong matching, $\mu > \mu_C$, delays that transition by raising the value of preserving future entry.

The intermediate region $\mu_G < \mu < \mu_E$ is especially informative. Buyer growth rises even though fewer sellers enter in period 2, because AI increases the network contribution of each remaining

seller. Thus, late-entry counts need not track buyer-side market expansion. The effect on early seller recruitment is also not globally monotone: stronger matching can raise early recruitment, but can reduce it when the installed base is small and forward-looking sellers place sufficiently high value on waiting for a more valuable future market. The proof below gives the exact thresholds.

The extension treats the AI state as exogenous and allows it to affect both support productivity and buyer-side matching or substitution. Endogenizing adoption would require an AI adoption cost. Additive capability and flexible seller transfers make support separable from network scale; this is a tractability benchmark, not a general investment-neutrality claim. Because the extension specifies neither an AI resource cost nor a social discount factor, the propositions concern participation and market development rather than welfare.

For this subsection only, set

$$s(a) = \alpha + \beta_0 + \mu a, \quad Z(a) = Q(a)^2 - 4\delta_s H(a) + 4\delta_p [3 - s(a)^2], \quad (\text{OA.4.40})$$

where $Q(a) = 4 - s(a)^2$ and $H(a) = 2 - s(a)^2$.

Proof of Proposition 4.4. The support first-order condition immediately gives (OA.4.38). With $\mu = 0$, only $\mathcal{V}(a)$ changes. Within the positive-late-entry region,

$$\frac{\partial n_1^{s*}}{\partial \mathcal{V}} = \frac{2Q(a)}{Z(a)} > 0, \quad (\text{OA.4.41})$$

$$\frac{\partial n_{2E}^{s*}}{\partial \mathcal{V}} = -\frac{2H(a)}{Z(a)} < 0, \quad (\text{OA.4.42})$$

$$\frac{\partial(n_1^{s*} + n_{2E}^{s*})}{\partial \mathcal{V}} = \frac{4}{Z(a)} > 0, \quad (\text{OA.4.43})$$

$$\frac{\partial(n_2^{b*} - n_1^{b*})}{\partial \mathcal{V}} = -\frac{s(a)H(a)}{Z(a)} < 0. \quad (\text{OA.4.44})$$

Multiplying by $\mathcal{V}'(a) > 0$ proves the proposition. The difference between $\mathcal{V}(a)$ and the positive-to-zero-late-entry boundary is increasing in a when $\mu = 0$, so AI moves the equilibrium toward zero late entry. $\square$

Proof of Proposition 4.5. Evaluate all derivatives at $a = 0$. Let

$$s_0 = \alpha + \beta_0, \quad Q_0 = 4 - s_0^2, \quad H_0 = 2 - s_0^2, \quad Z_0 = Q_0^2 - 4\delta_s H_0 + 4\delta_p (3 - s_0^2). \quad (\text{OA.4.45})$$

Because $s(0) = s_0$ does not depend on μ , the marginal AI effect is affine in μ . In this proof only, a subscript s denotes a derivative with respect to the scalar network index $s(a)$, evaluated at s_0 while $\mathcal{V}$ is held fixed; it does not denote a seller cohort. Define

$$A_0 \equiv \frac{2Q_0}{Z_0} \mathcal{V}'(0) > 0. \quad (\text{OA.4.46})$$

Then

$$\left. \frac{dn_1^{s*}}{da} \right|_{a=0} = A_0 + \mu n_{1,s}^{s*}, \quad (\text{OA.4.47})$$

$$\left. \frac{dn_{2E}^{s*}}{da} \right|_{a=0} = -\frac{H_0}{Q_0} A_0 + \mu n_{2E,s}^{s*}, \quad (\text{OA.4.48})$$

$$\left. \frac{d(n_1^{s*} + n_{2E}^{s*})}{da} \right|_{a=0} = \frac{2}{Q_0} A_0 + \mu M_s, \quad (\text{OA.4.49})$$

$$\left. \frac{d(n_2^{b*} - n_1^{b*})}{da} \right|_{a=0} = -\frac{s_0 H_0}{2Q_0} A_0 + \mu G_s. \quad (\text{OA.4.50})$$

Sign verification.

To establish the signs used below, set $y = s^2$, $Q = 4 - y$, $H = 2 - y$, and $Z = Q^2 - 4\delta_s H + 4\delta_p(3 - y)$; suppressing the argument s , define

$$R \equiv Q + 2(\delta_p - \delta_s), \quad B \equiv 1 + \delta_p - \delta_s \frac{H}{Q}, \quad C \equiv 6 - y + 4\delta_p - 4\delta_s \frac{3 - y}{Q}.$$

Then $Z_y = -2R$, $B_y = 2\delta_s/Q^2$, and $C_y = -1 + 4\delta_s/Q^2$. The positive-late-entry formulas can be written as

$$n_{2E}^{s*} = \frac{2sB}{Z} - \frac{2H\mathcal{V}}{Z}, \quad n_1^{s*} + n_{2E}^{s*} = \frac{sC}{Z} + \frac{4\mathcal{V}}{Z}, \quad n_2^{b*} - n_1^{b*} = \frac{sn_{2E}^{s*}}{2}.$$

Consequently, evaluated at $s = s_0$ and $\mathcal{V} = \mathcal{V}(0)$,

$$\begin{aligned} n_{2E,s}^{s*} &= \frac{2\{Z_0(B_0 + 2s_0^2 B_{y,0}) + 4s_0^2 R_0 B_0\} + 4s_0 \mathcal{V}(0)(Q_0 s_0^2 + 4\delta_p)}{Z_0^2} > 0, \\ M_s &= \frac{Z_0(C_0 + 2s_0^2 C_{y,0}) + 4s_0^2 R_0 C_0 + 16s_0 \mathcal{V}(0) R_0}{Z_0^2} > 0, \\ G_s &= \frac{n_{2E}^{s*} + s_0 n_{2E,s}^{s*}}{2} > 0. \end{aligned}$$

Indeed, $Z_0 > 0$, $R_0 > Q_0 - 2 > 0$, $B_0 > 1 - H_0/Q_0 > 0$, and $B_{y,0} > 0$. Moreover,

$$C_0 > \frac{12 - 6s_0^2 + s_0^4}{Q_0} > 0$$

and, writing $y_0 = s_0^2$,

$$C_0 + 2y_0 C_{y,0} > \frac{48 - 60y_0 + 26y_0^2 - 3y_0^3}{Q_0^2} > 0,$$

where the last numerator is decreasing on $[0, 1]$ and equals 11 at $y_0 = 1$. These inequalities establish the displayed signs. The positivity of $n_{2E,s}^{s*}$ additionally uses $Z - 2HR = Qs^2 + 4\delta_p > 0$. Therefore, define

$$\mu_M \equiv -\frac{2A_0}{Q_0 M_s} < 0, \quad (\text{OA.4.51})$$

$$\mu_G \equiv \frac{s_0 H_0 A_0}{2Q_0 G_s} > 0, \quad (\text{OA.4.52})$$

$$\mu_E \equiv \frac{H_0 A_0}{Q_0 n_{2E,s}^{s*}} > 0. \quad (\text{OA.4.53})$$

These definitions yield the three threshold equivalences in Proposition 4.5. Moreover,

$$\frac{\mu_G}{\mu_E} = \frac{s_0 n_{2E,s}^{s*}}{n_{2E}^{s*} + s_0 n_{2E,s}^{s*}} < 1, \quad (\text{OA.4.54})$$

which proves $0 < \mu_G < \mu_E$.

Finally, define the gap between net capability value and the positive-to-zero-late-entry boundary by

$$\Gamma(a) = \mathcal{V}(a) - \left[\frac{(1 + \delta_p)s(a)}{H(a)} - \frac{\delta_s s(a)}{Q(a)} \right]. \quad (\text{OA.4.55})$$

At the baseline,

$$\Gamma'(0) = \mathcal{V}'(0) - \mu T(s_0), \quad (\text{OA.4.56})$$

where

$$T(s_0) = (1 + \delta_p) \frac{2 + s_0^2}{H_0^2} - \delta_s \frac{4 + s_0^2}{Q_0^2} > 0. \quad (\text{OA.4.57})$$

Thus

$$\mu_C \equiv \frac{\mathcal{V}'(0)}{T(s_0)} > 0, \quad \Gamma'(0) > 0 \iff \mu < \mu_C, \quad (\text{OA.4.58})$$

which gives the zero-late-entry transition result. $\square$

Early-recruitment response.

At the baseline, the network-index derivative of early recruitment can change sign. Explicitly,

$$n_{1,s}^{s*} = \frac{2Q_0}{Z_0} \Xi_0, \quad (\text{OA.4.59})$$

where

$$\Xi_0 = \frac{1}{2} + s_0 n_1^{s*} - \delta_s \frac{4 + s_0^2 + 8s_0 n_1^{s*}}{Q_0^2} + \delta_p \frac{4 + s_0^2 + 4s_0 n_1^{s*}}{Q_0^2}. \quad (\text{OA.4.60})$$

This term can be negative when sellers are sufficiently patient relative to the platform and the installed base is small. Proposition 4.5 therefore does not impose an unconditional sign on early recruitment.